



# 自然语言处理

## 8. Transformer与预训练模型

虞剑飞

南京大学智能科学与技术学院

2025.4.30

# 本章内容

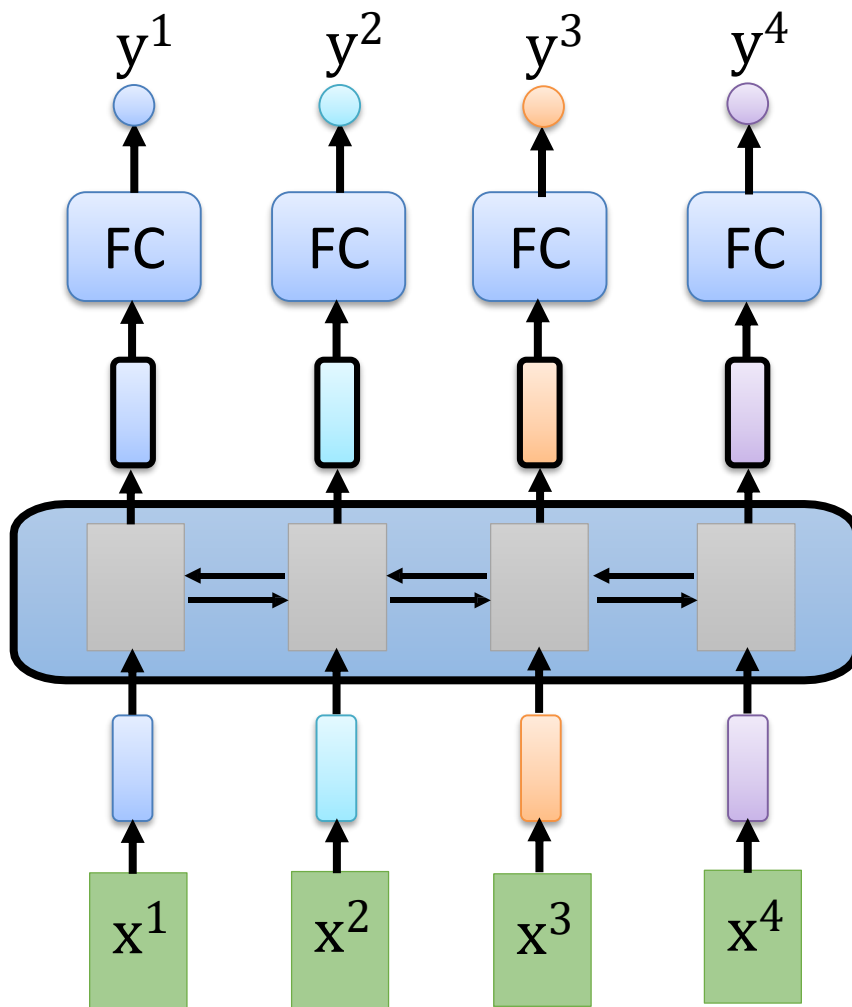
- 8.1 引言
- 8.2 自注意力机制
- 8.3 Transformer网络架构
- 8.4 预训练模型及其应用
  - 8.4.1 基于Transformer编码器的预训练语言模型
  - 8.4.2 基于Transformer的预训练Seq2Seq模型
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- 8.5 本章小结

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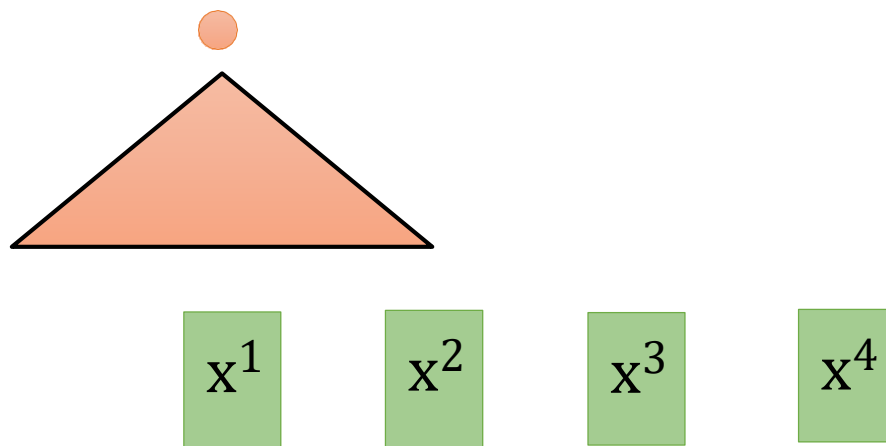
# 引言

- 循环神经网络缺陷
  - 很难并行化



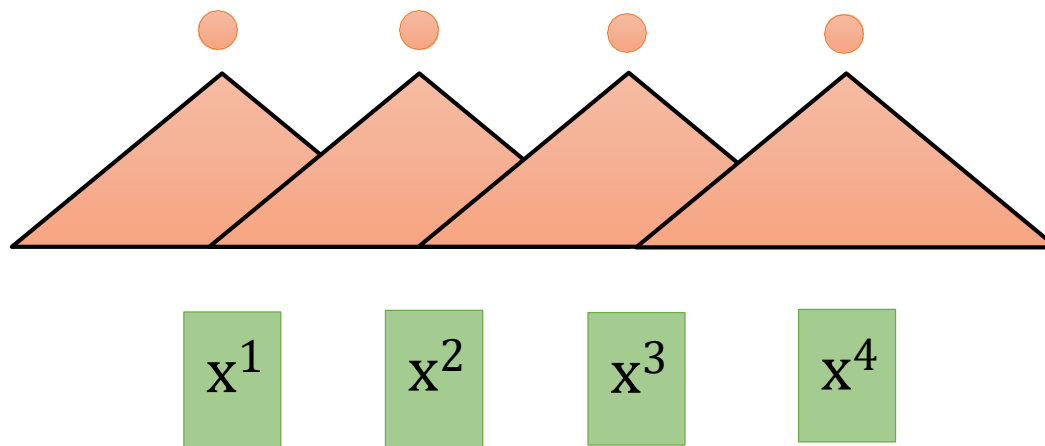
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- 循环神经网络缺陷
  - 很难并行化
    - 方案1：使用CNN来替换RNN



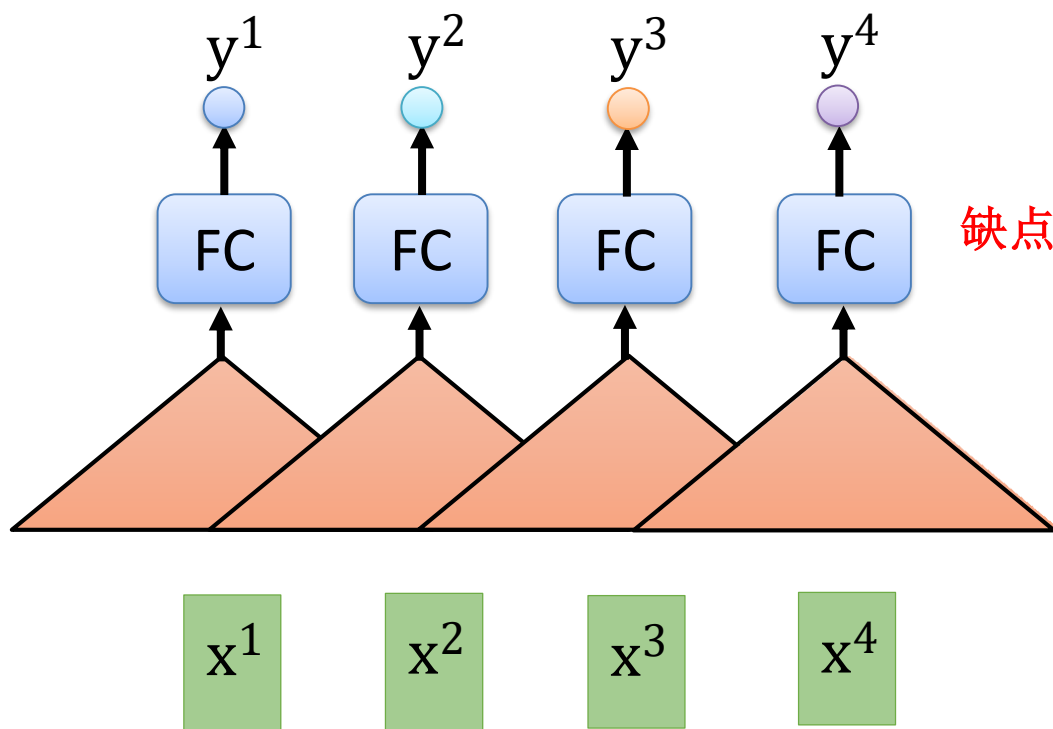
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# 引言

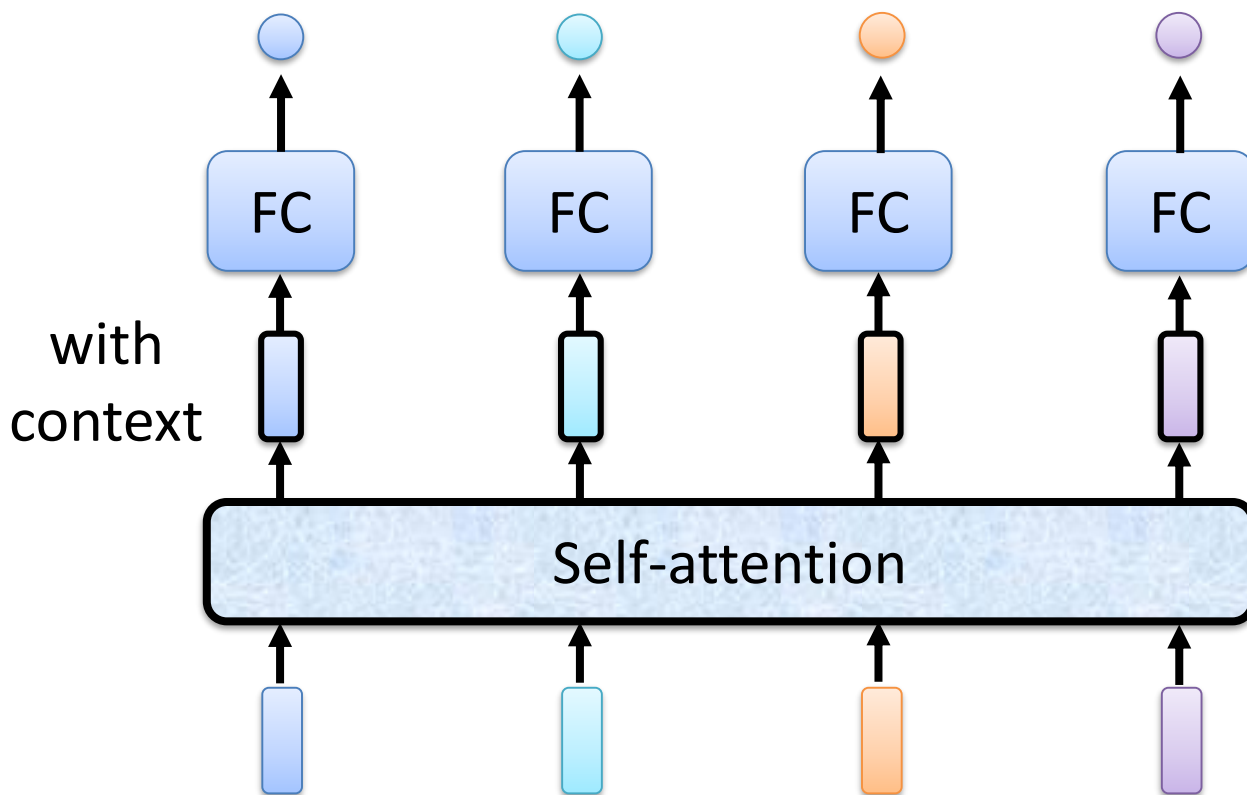
- 循环神经网络缺陷
  - 很难并行化
    - 方案1：使用CNN来替换RNN



缺点：只能捕获局部信息

# 引言

- 循环神经网络缺陷
  - 很难并行化
    - 方案2：自注意力机制（Self-Attention Mechanism）



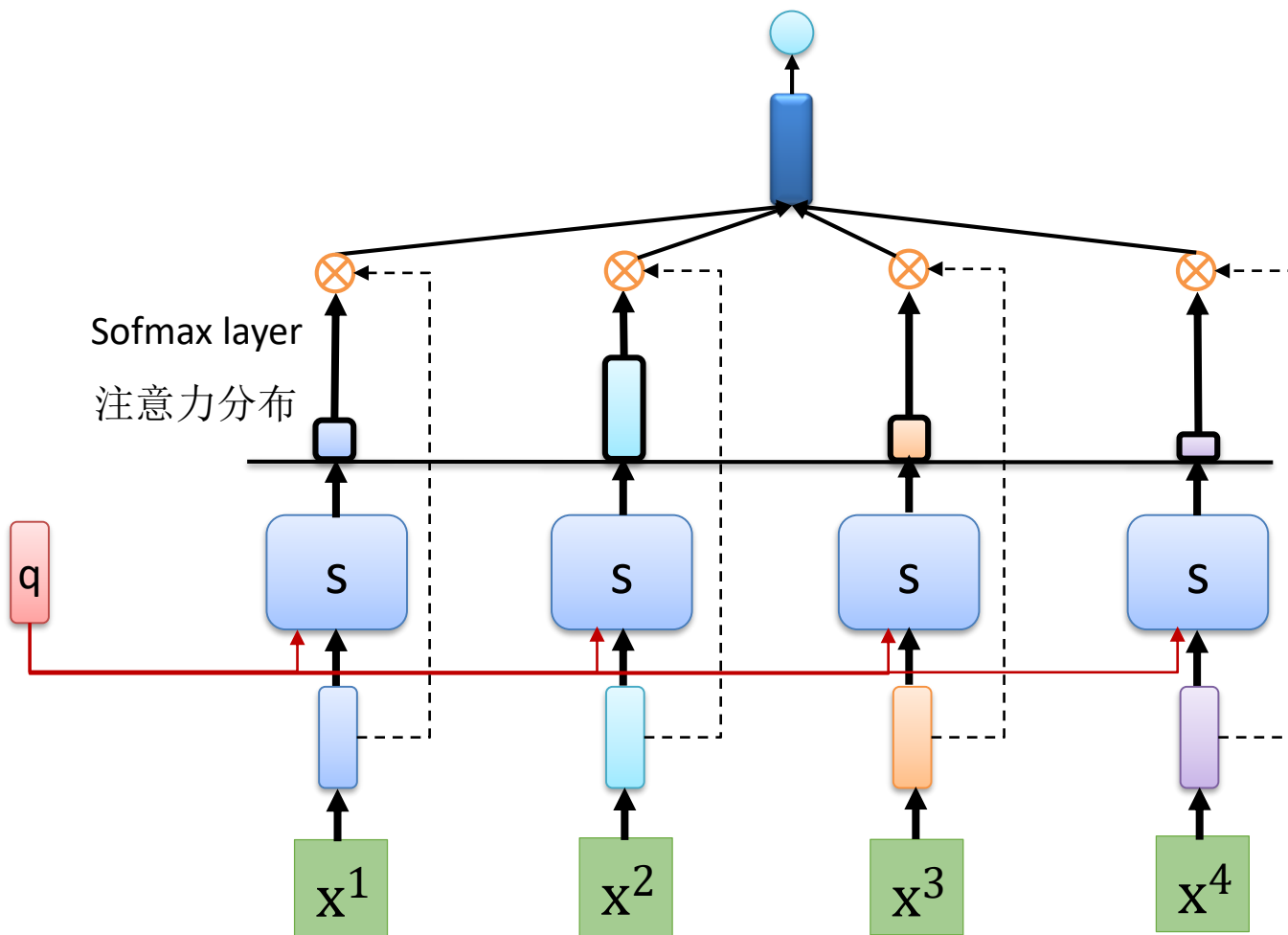


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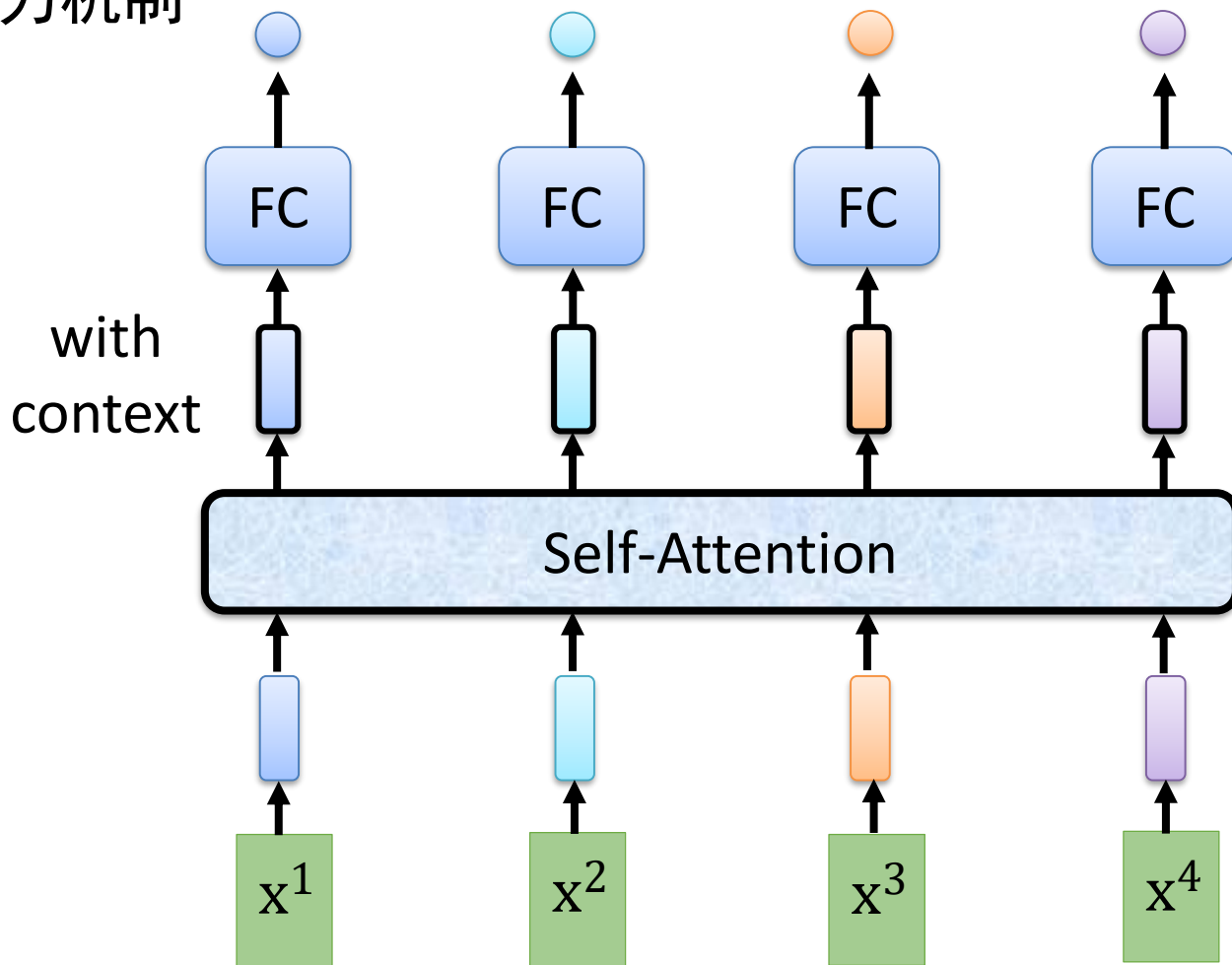
# 自注意力机制 (Self-Attention)

- 注意力机制回顾



# 自注意力机制 (Self-Attention)

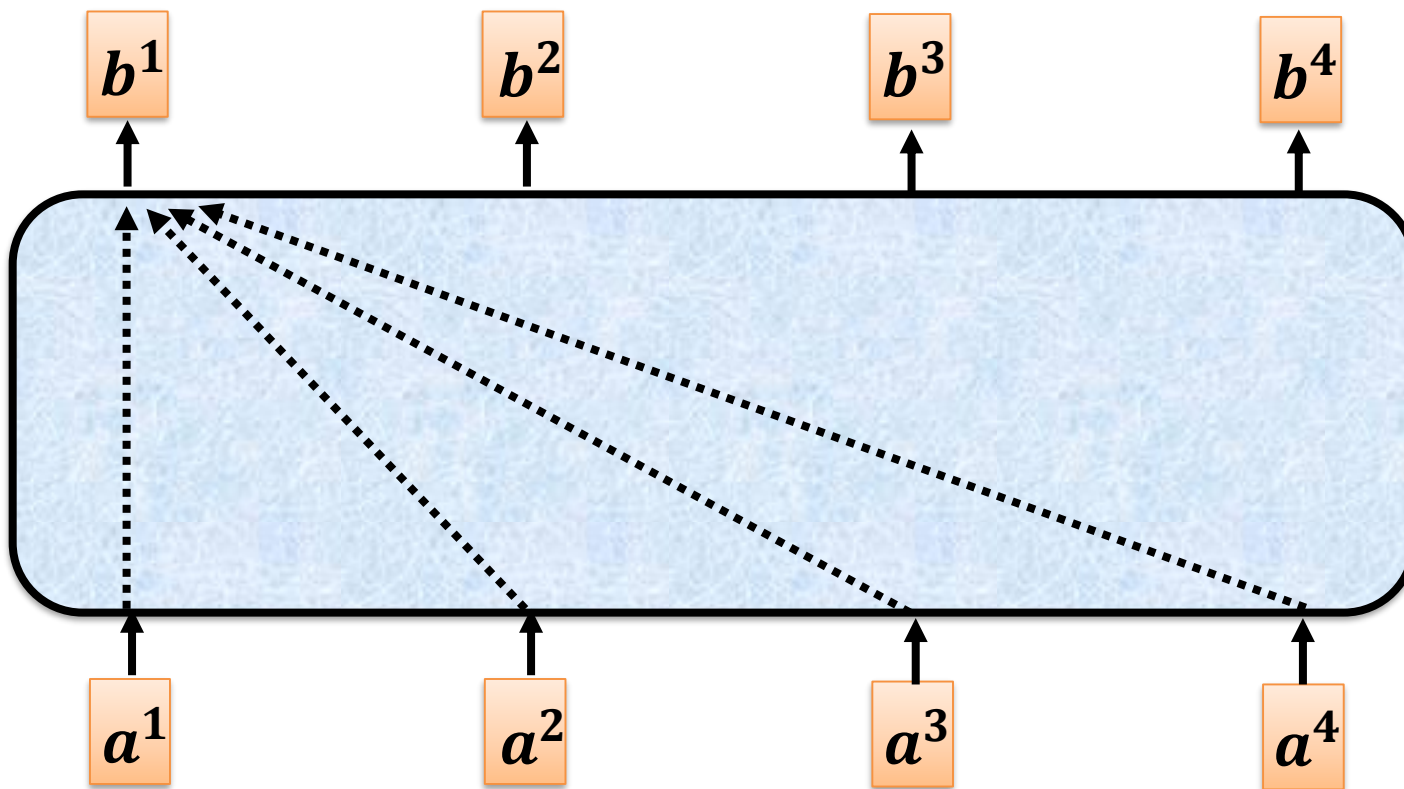
- 自注意力机制



Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I.. Attention is all you need. NIPS 2017.

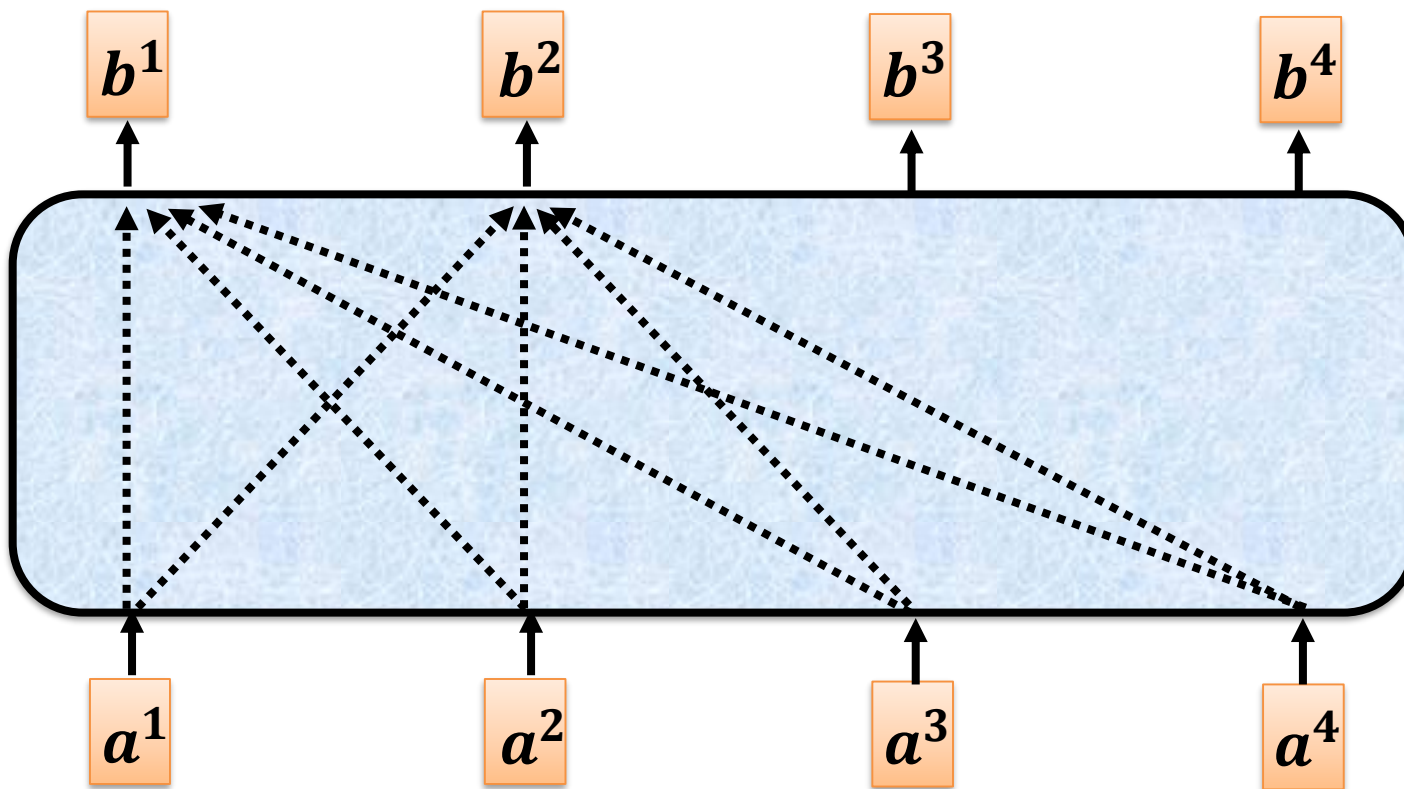
# 自注意力机制 (Self-Attention)

- 自注意力机制



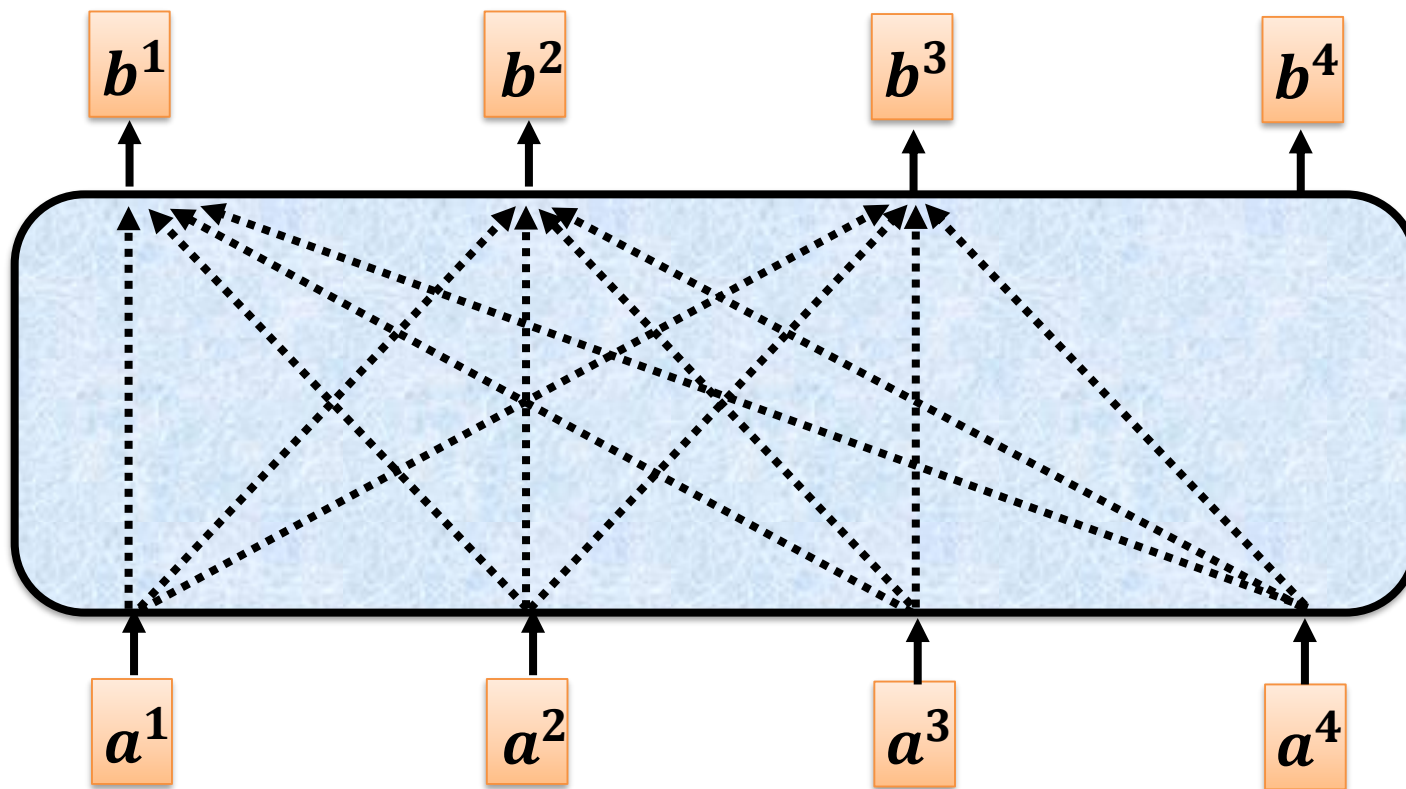
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- 自注意力机制



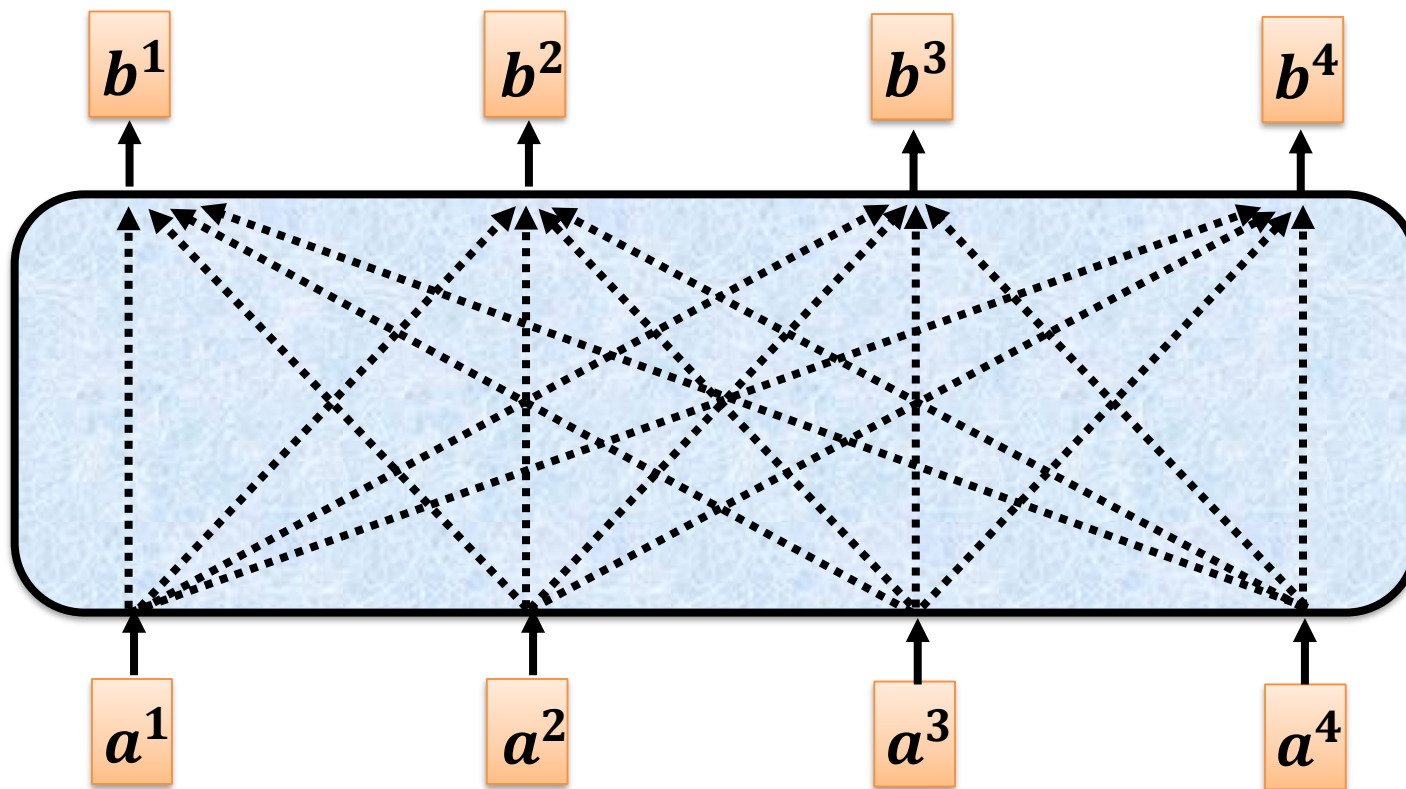
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- 自注意力机制



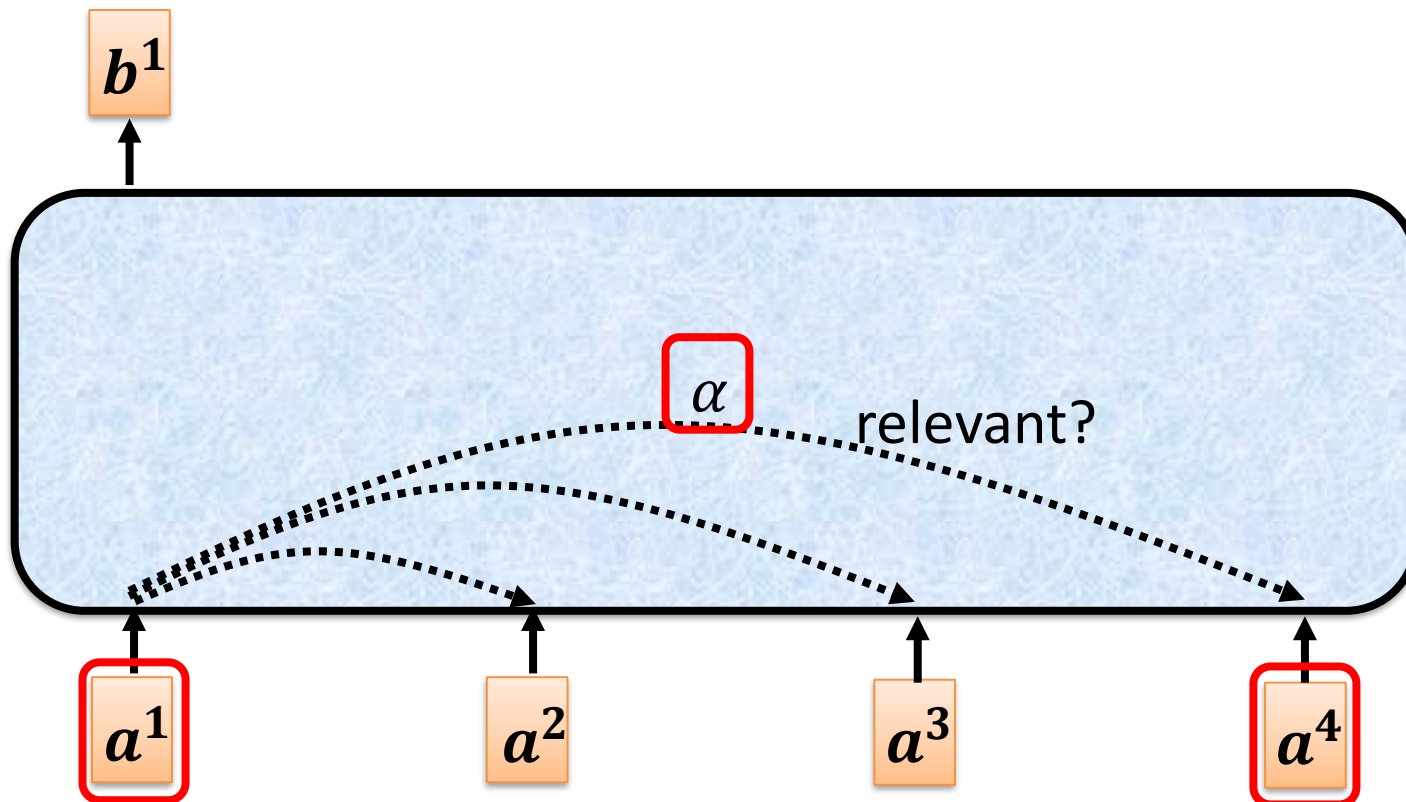
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- 自注意力机制



# 自注意力机制 (Self-Attention)

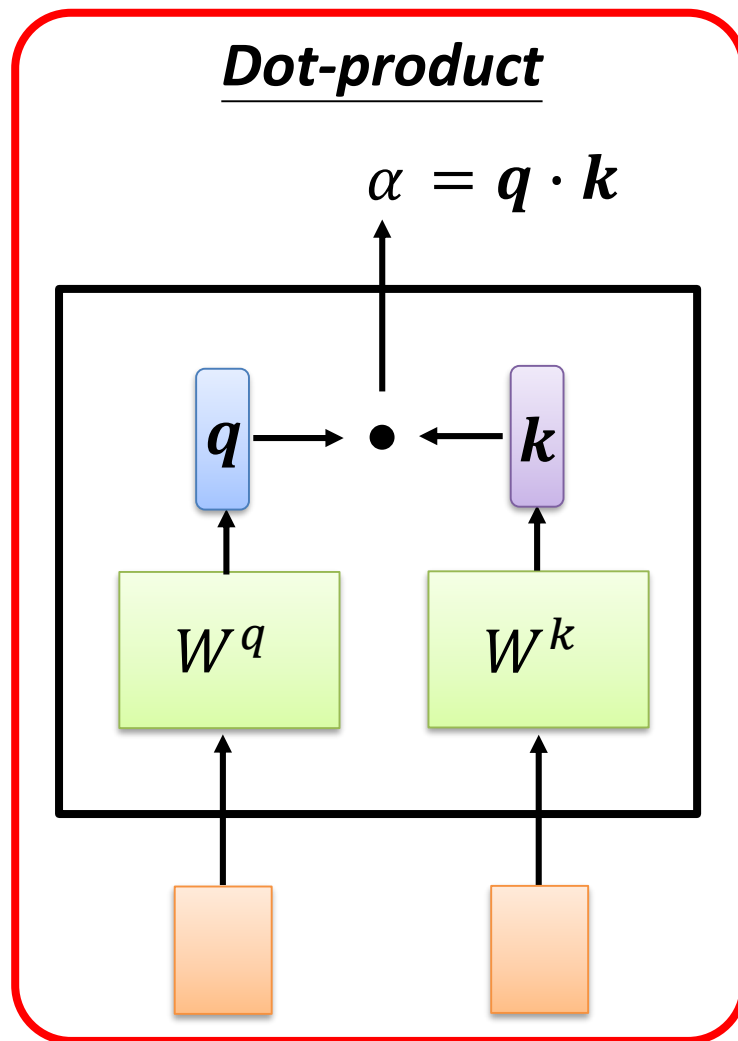
- 自注意力机制





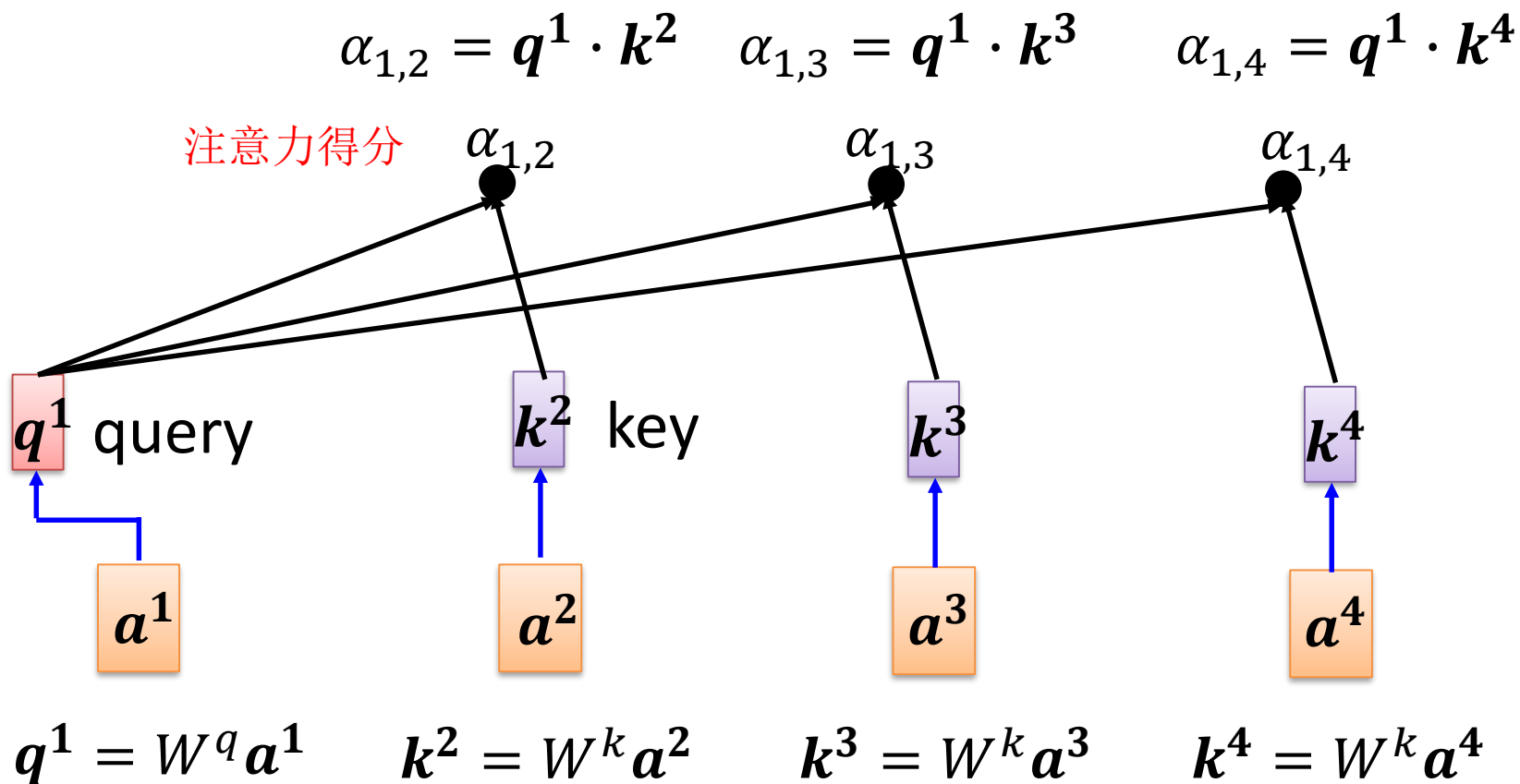
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- 自注意力机制



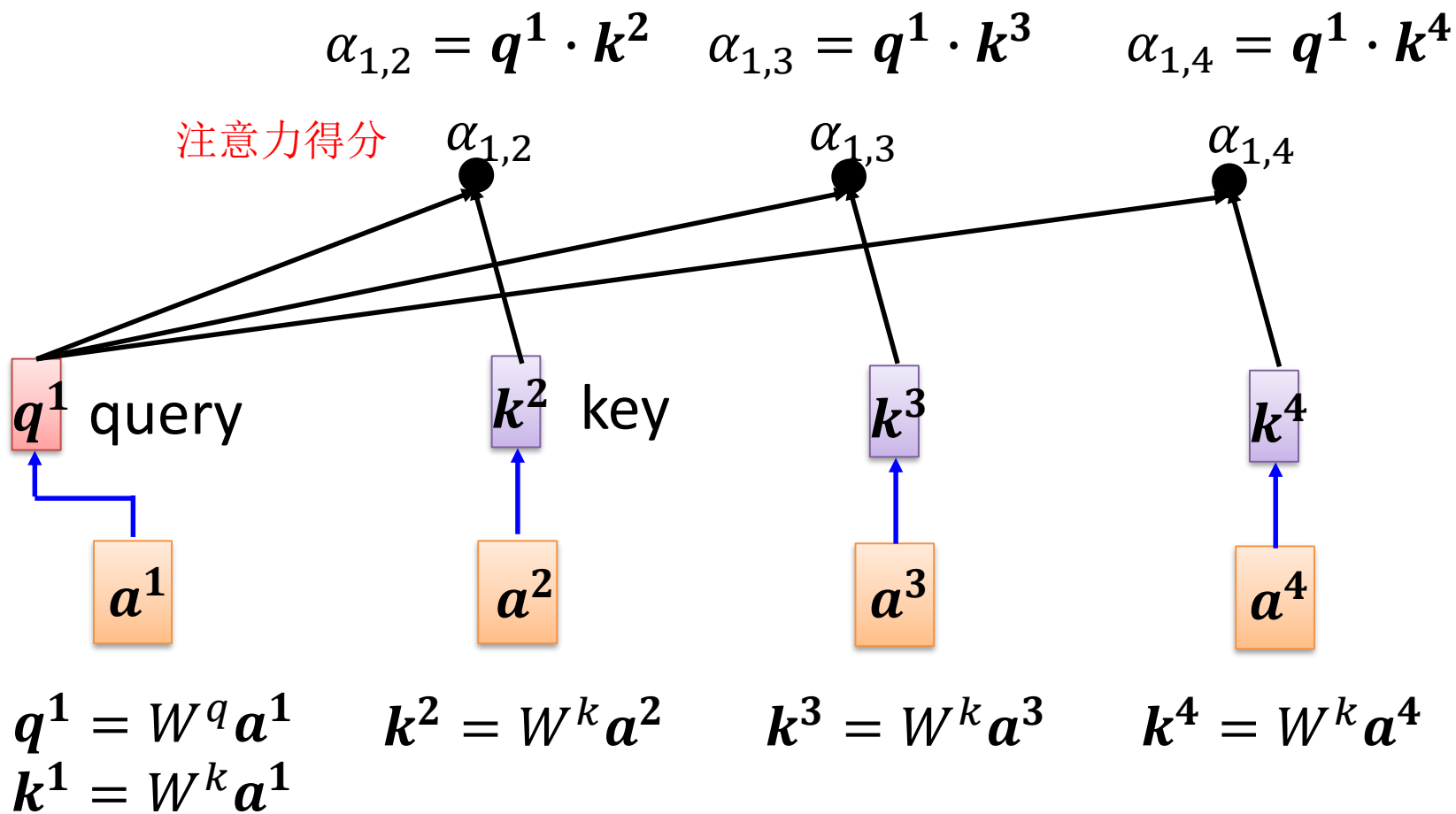
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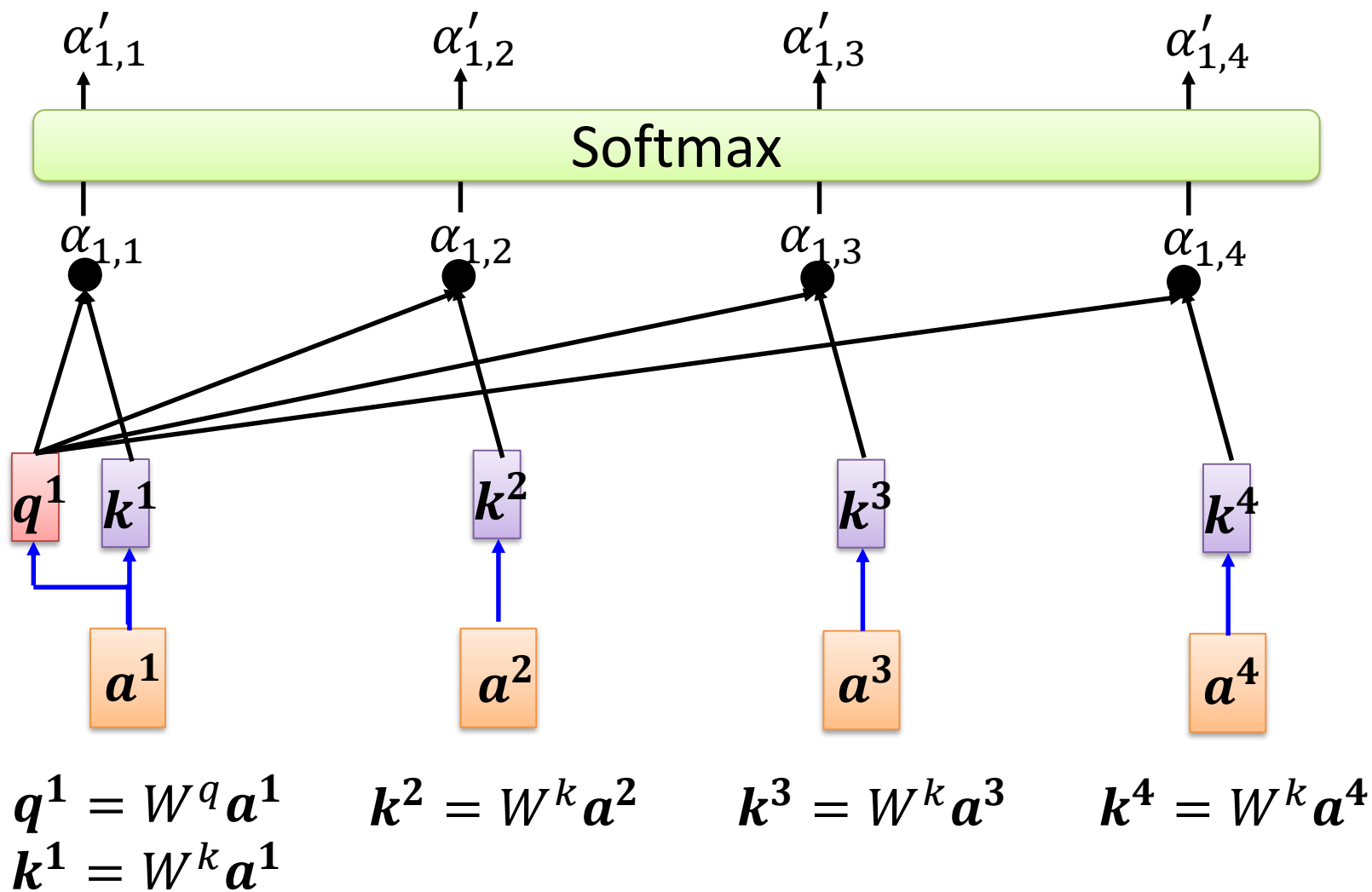
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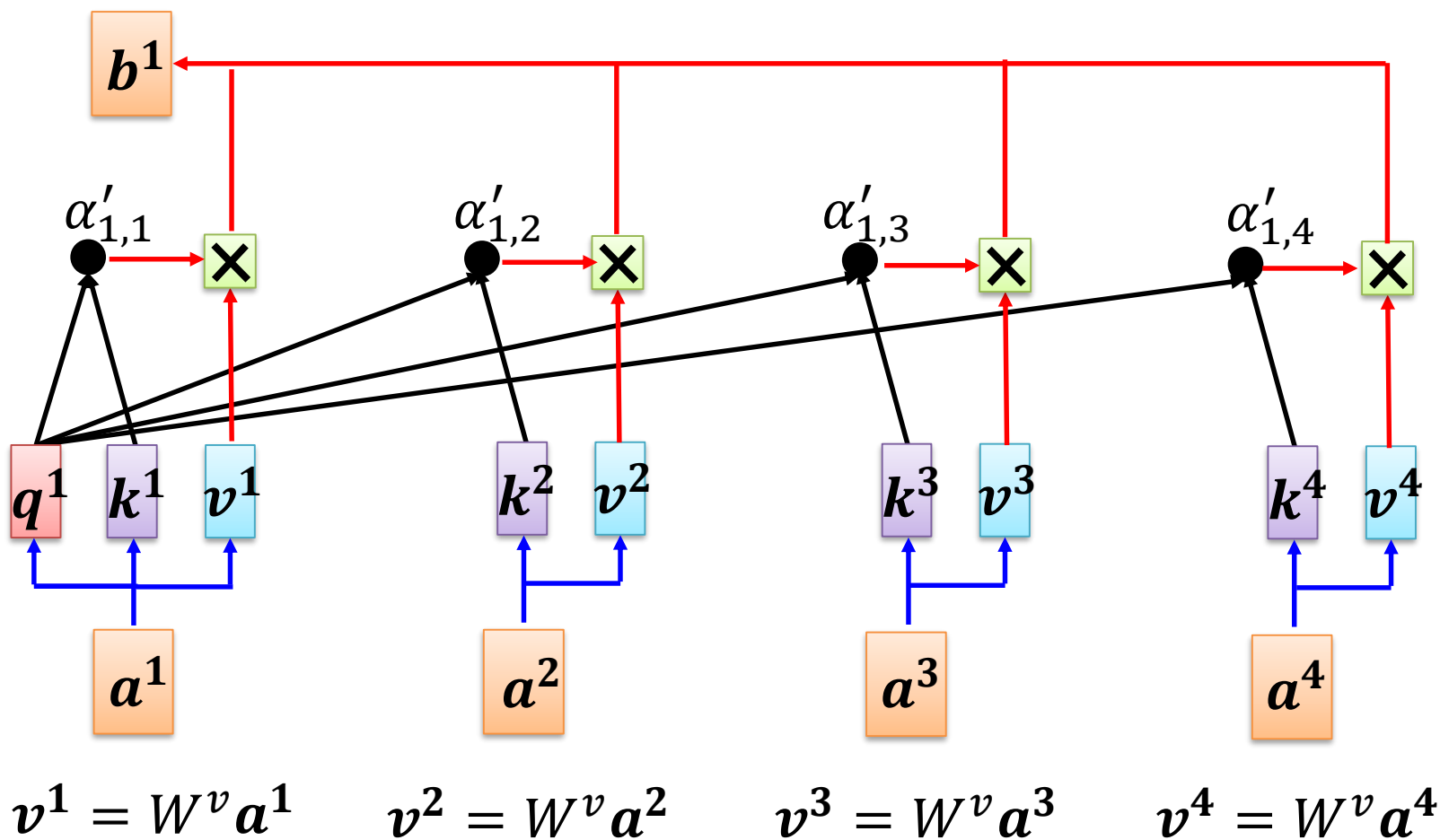
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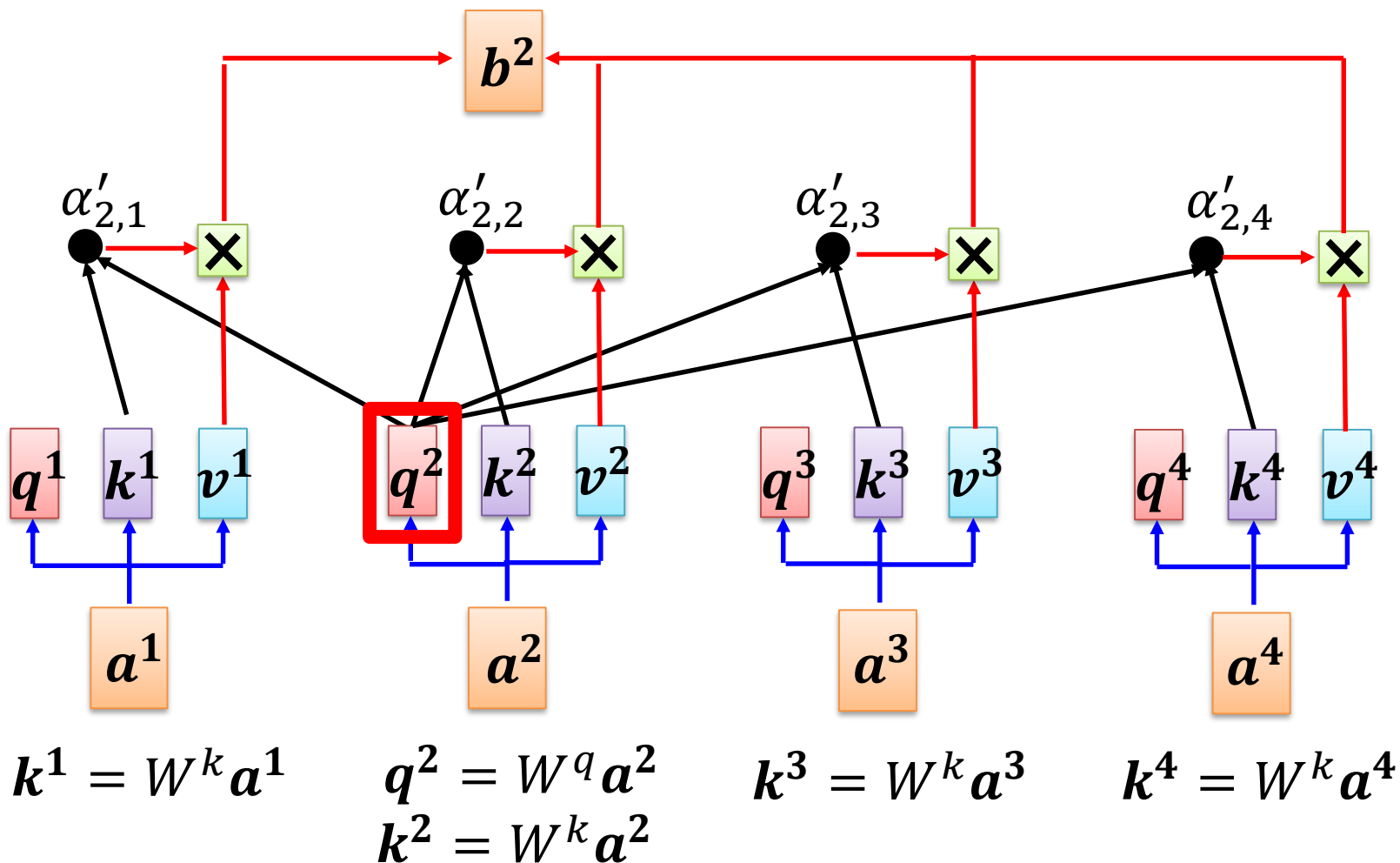
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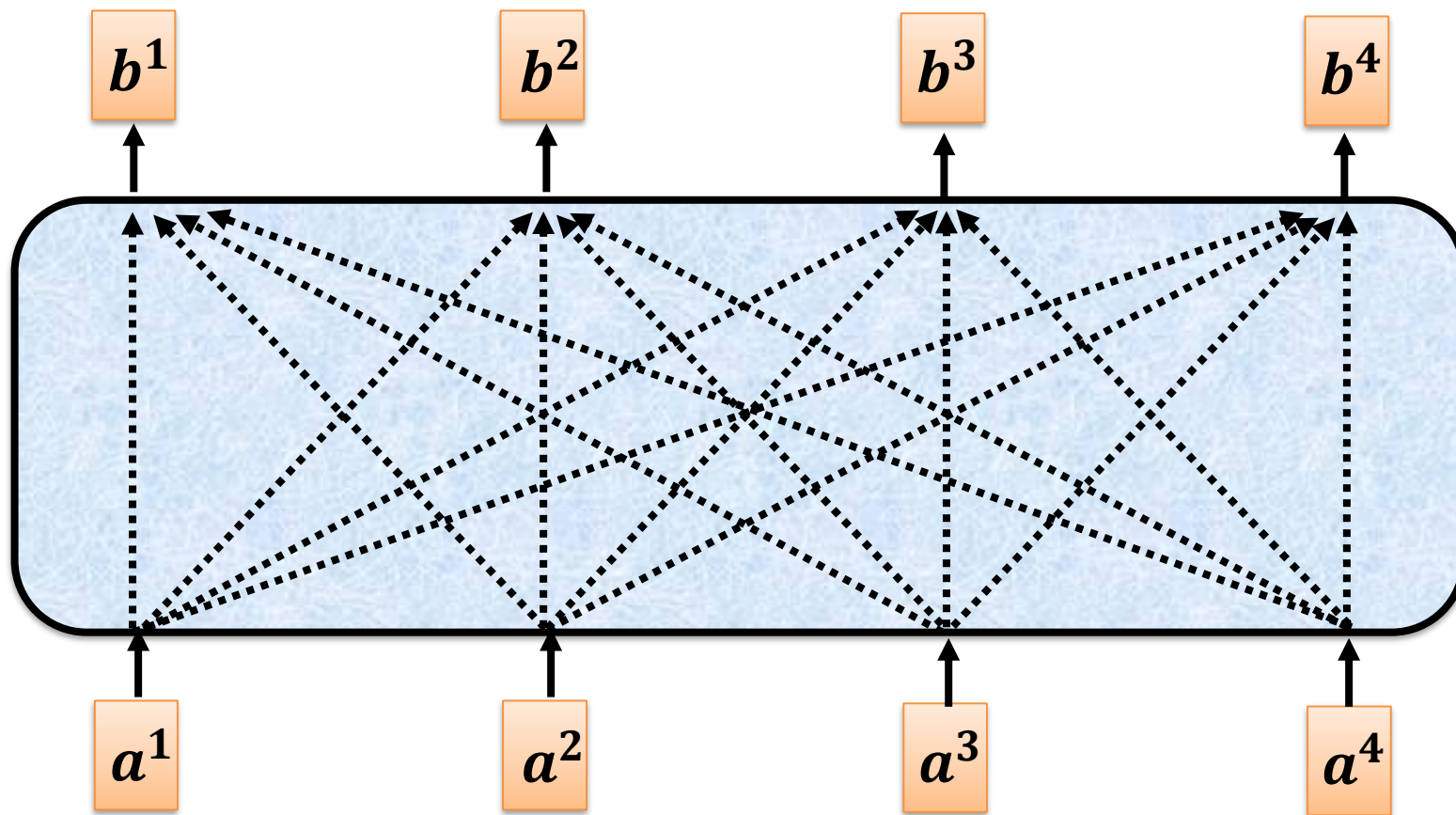
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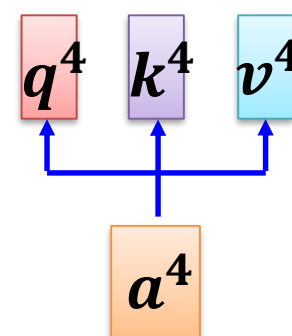
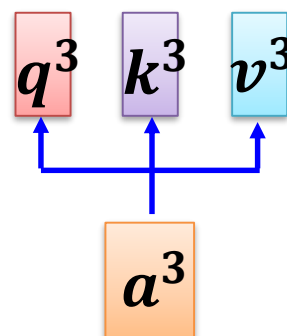
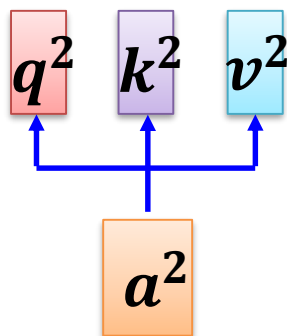
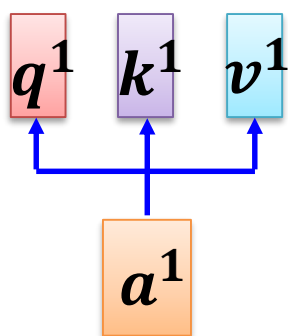
# 自注意力机制 (Self-Attention)

- 自注意力机制

$$q^i = W^q a^i \quad \begin{matrix} q^1 & q^2 & q^3 & q^4 \\ Q \end{matrix} = \begin{matrix} W^q \\ I \end{matrix} \begin{matrix} a^1 & a^2 & a^3 & a^4 \end{matrix}$$

$$k^i = W^k a^i \quad \begin{matrix} k^1 & k^2 & k^3 & k^4 \\ K \end{matrix} = \begin{matrix} W^k \\ I \end{matrix} \begin{matrix} a^1 & a^2 & a^3 & a^4 \end{matrix}$$

$$v^i = W^v a^i \quad \begin{matrix} v^1 & v^2 & v^3 & v^4 \\ V \end{matrix} = \begin{matrix} W^v \\ I \end{matrix} \begin{matrix} a^1 & a^2 & a^3 & a^4 \end{matrix}$$

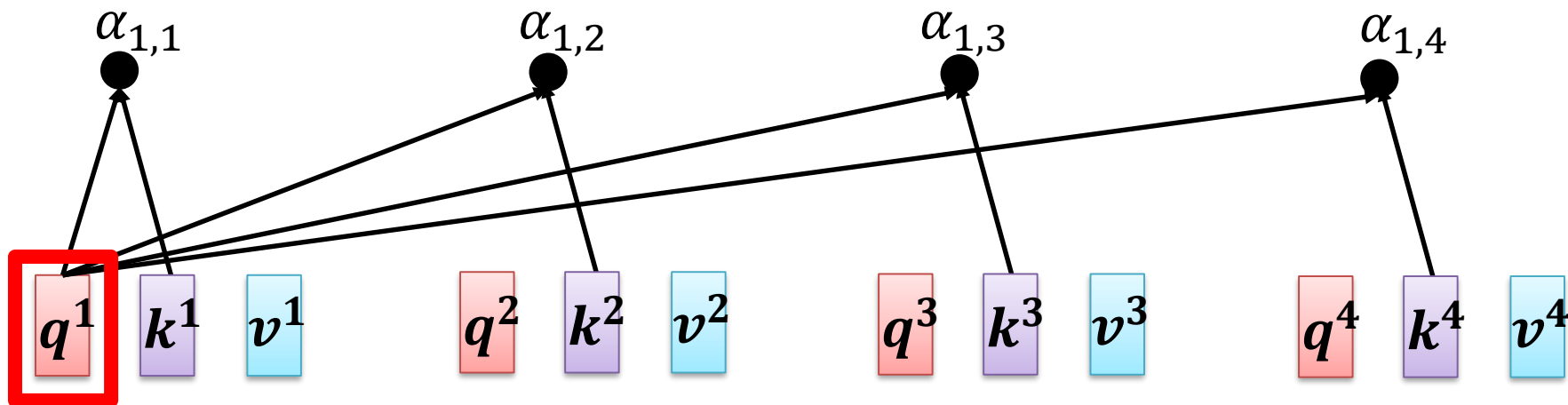




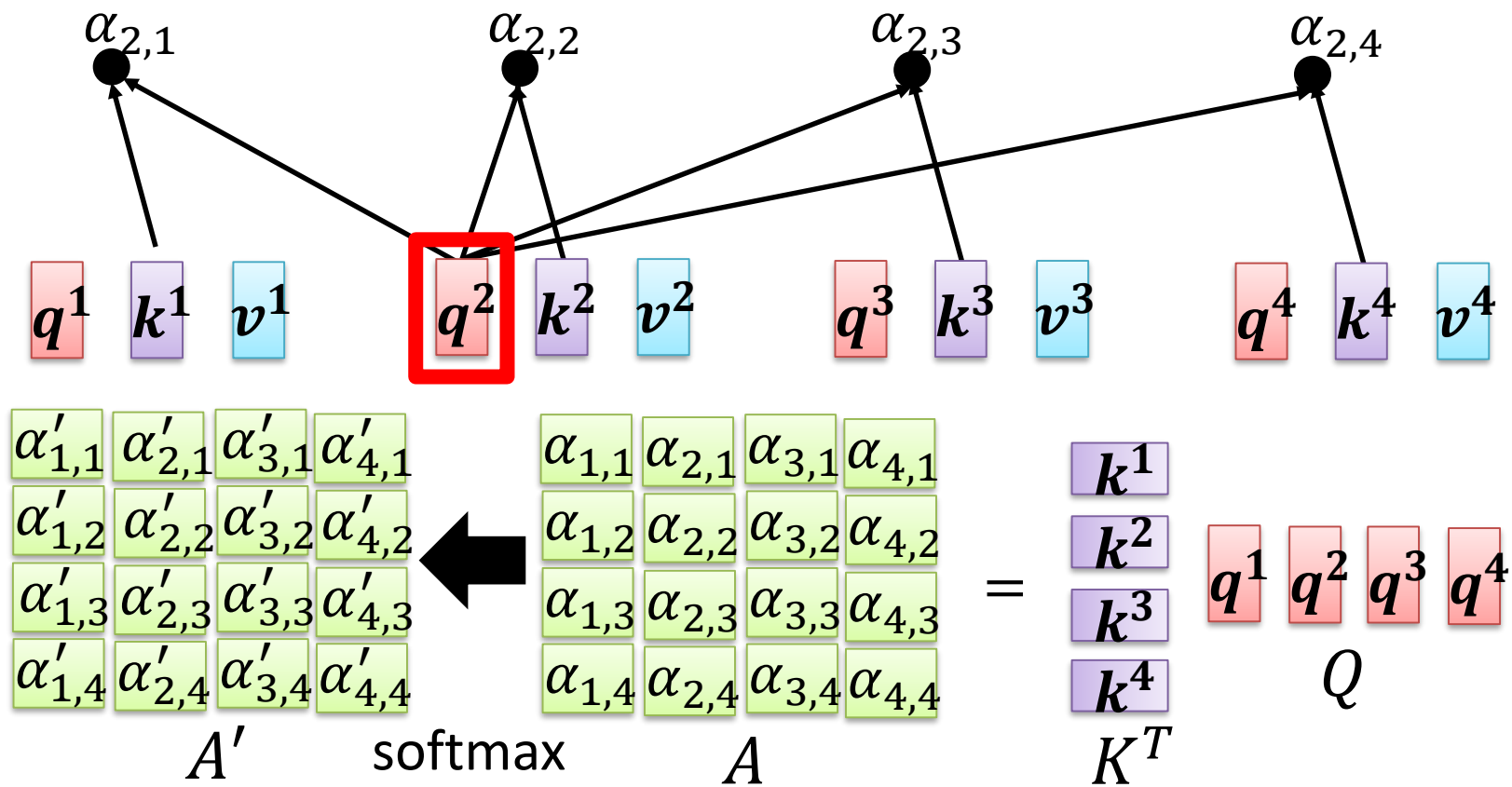
# 自注意力机制 (Self-Attention)

- 自注意力机制

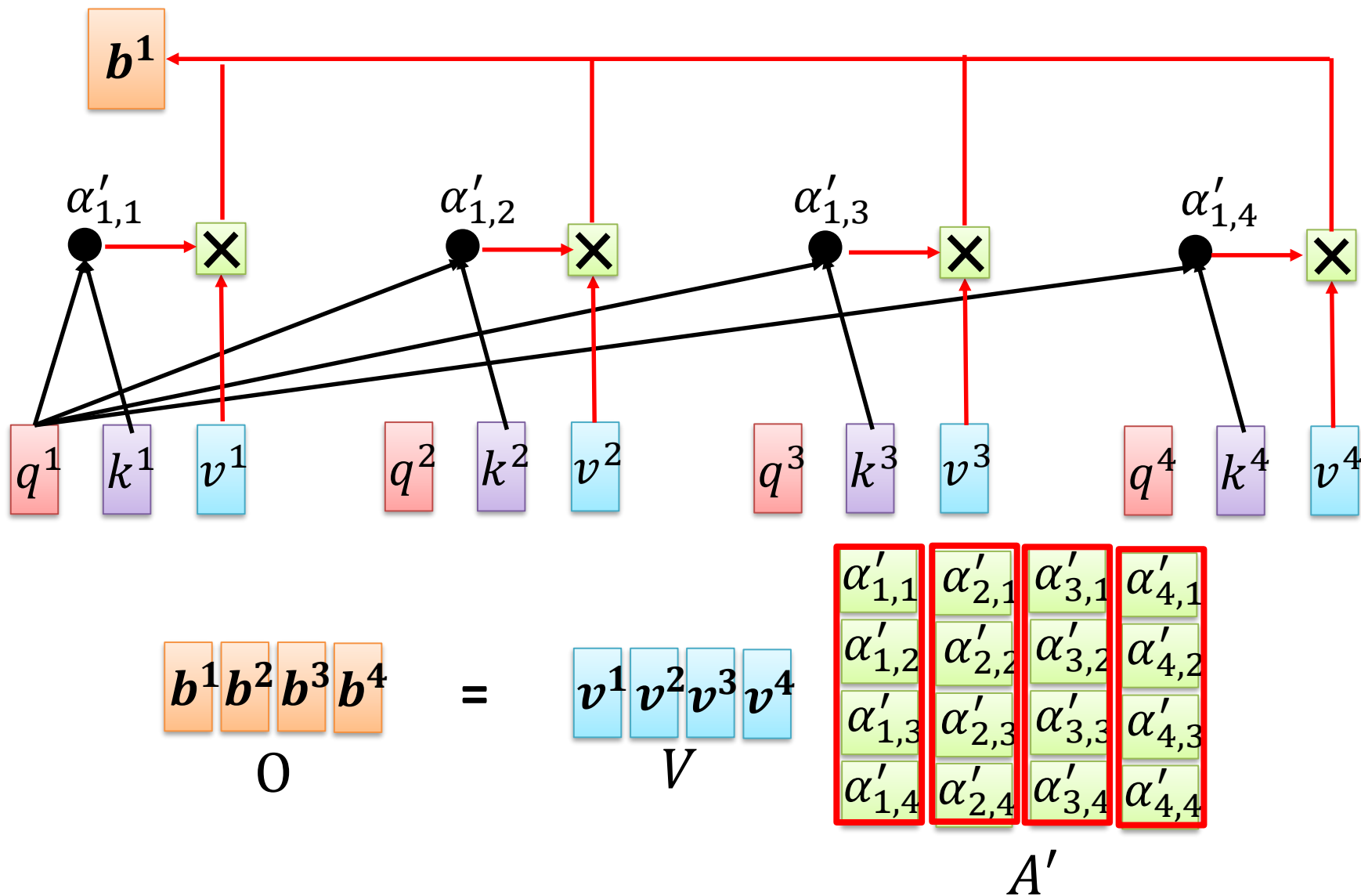
$$\begin{aligned} \alpha_{1,1} &= k^1 q^1 & \alpha_{1,2} &= k^2 q^1 \\ \alpha_{1,3} &= k^3 q^1 & \alpha_{1,4} &= k^4 q^1 \end{aligned}$$
$$\begin{matrix} \alpha_{1,1} \\ \alpha_{1,2} \\ \alpha_{1,3} \\ \alpha_{1,4} \end{matrix} = \begin{matrix} k^1 \\ k^2 \\ k^3 \\ k^4 \end{matrix} q^1$$



# 自注意力机制 (Self-Attention)



# 自注意力机制 (Self-Attention)



# 自注意力机制 (Self-Attention)

$$\begin{aligned} Q &= W^q I \\ K &= W^k I \\ V &= W^v I \end{aligned}$$

待学习参数

$$A' \leftarrow A = K^T Q$$

Attention Matrix

$$O = V A'$$

# 自注意力机制 (Self-Attention)

- 课堂练习

|   |
|---|
| 1 |
| 0 |
| 1 |
| 0 |

$x^1$

|   |
|---|
| 0 |
| 2 |
| 0 |
| 2 |

$x^2$

|   |
|---|
| 1 |
| 1 |
| 1 |
| 1 |

$x^3$

I

=

|   |   |   |
|---|---|---|
| 1 | 0 | 1 |
| 0 | 2 | 1 |
| 1 | 0 | 1 |
| 0 | 2 | 1 |

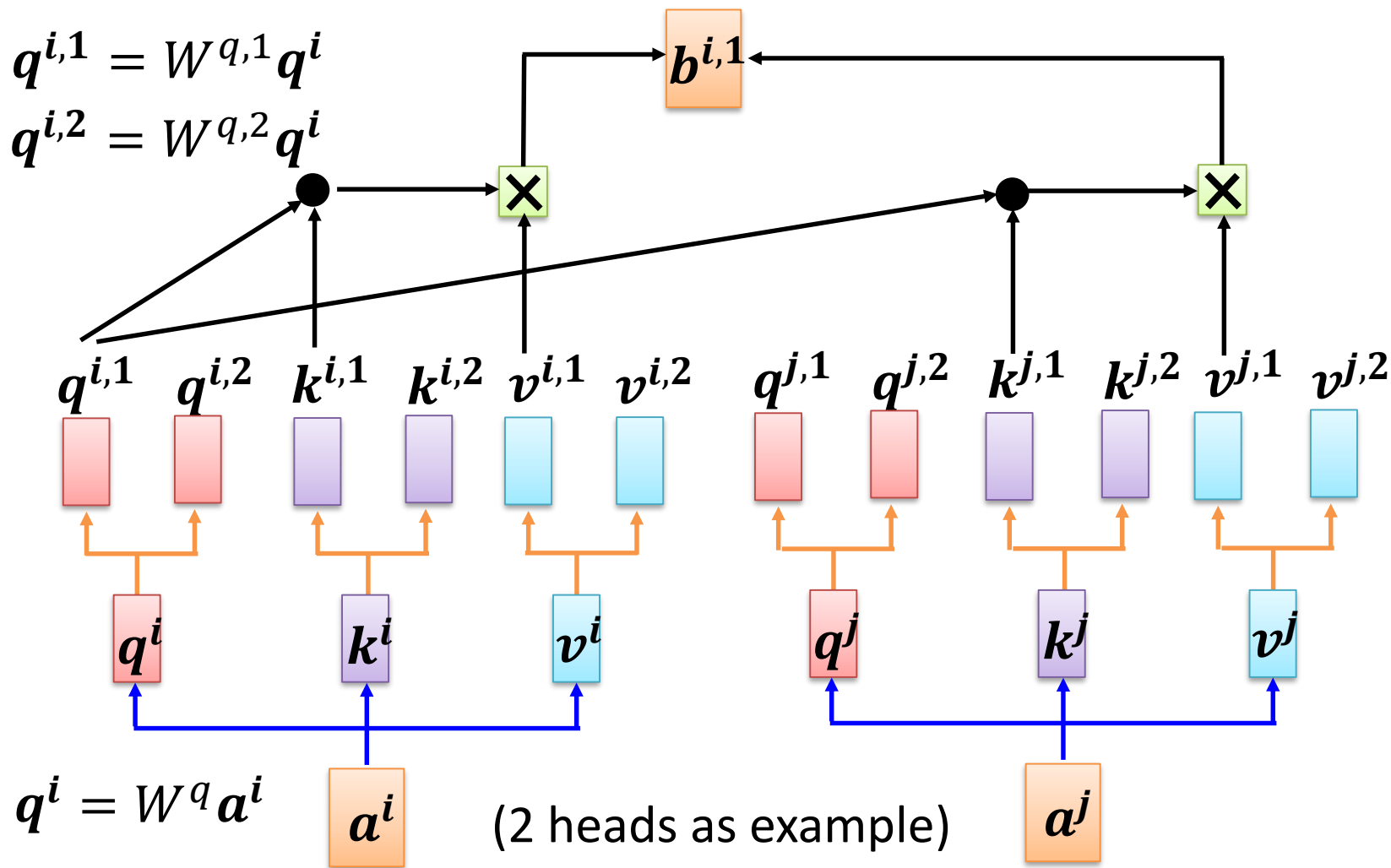
$$W^q = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$W^k = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$W^v = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 2 & 3 & 0 & 1 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

# 自注意力机制 (Self-Attention)

- 多头自注意力机制 (Multi-Head Self-Attention)

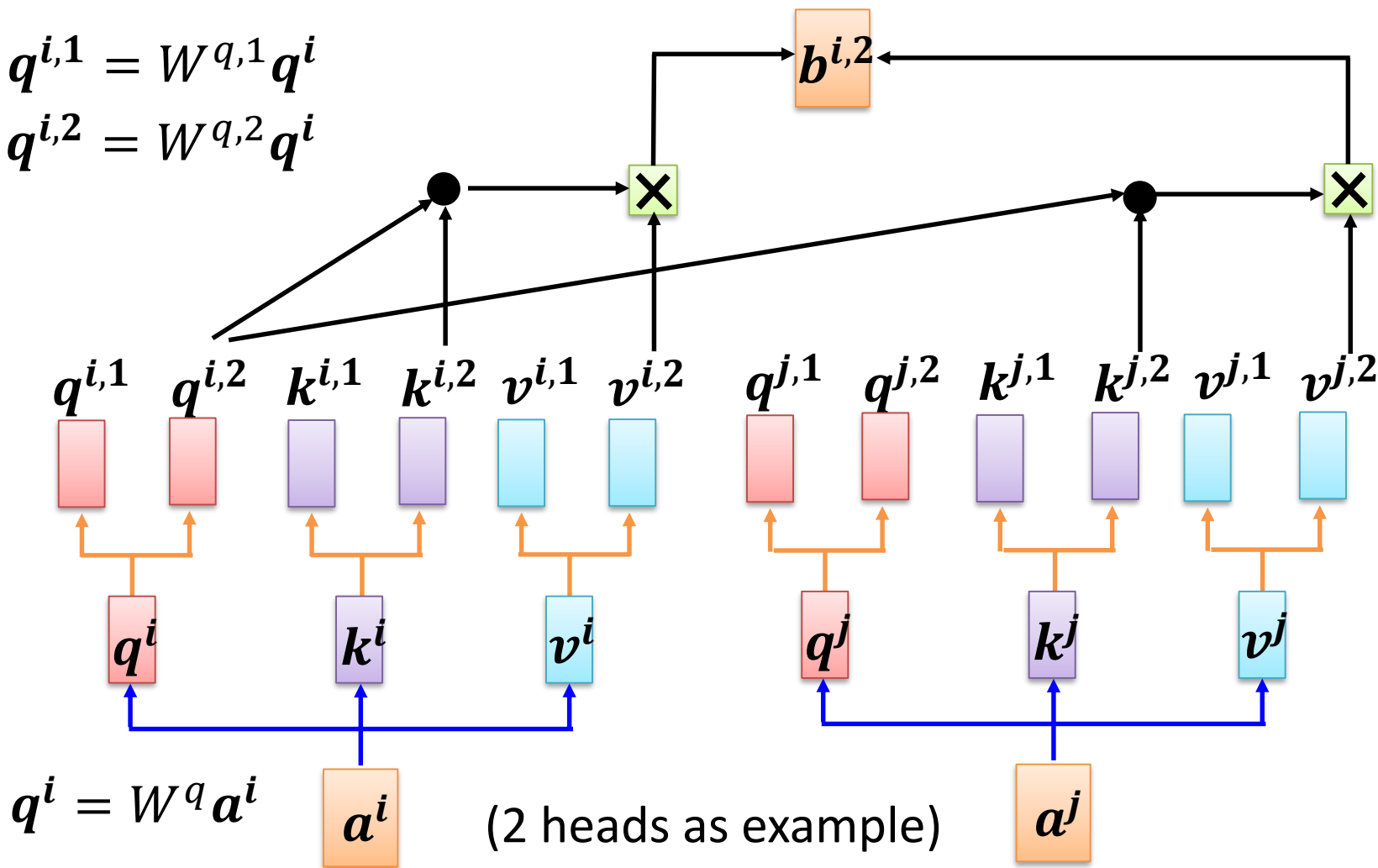


# 自注意力机制 (Self-Attention)

- 多头自注意力机制 (Multi-Head Self-Attention)

$$q^{i,1} = W^{q,1} q^i$$

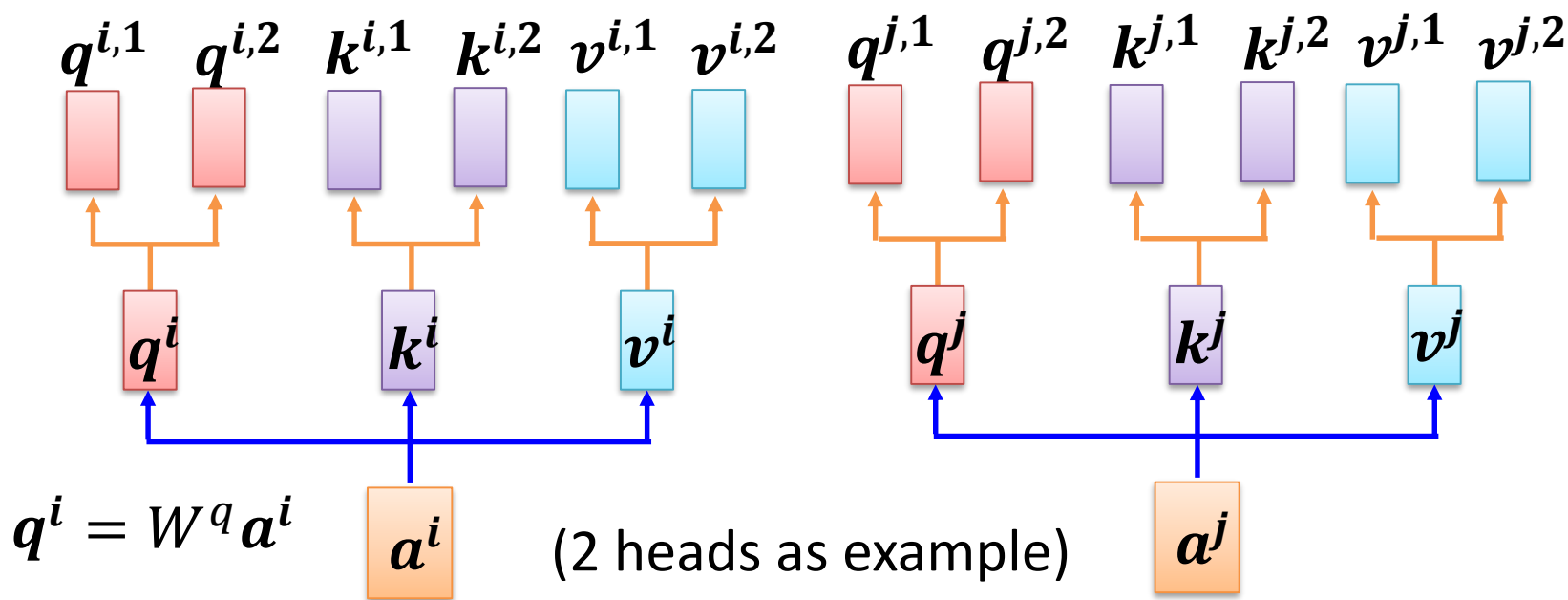
$$q^{i,2} = W^{q,2} q^i$$



# 自注意力机制 (Self-Attention)

- 多头自注意力机制 (Multi-Head Self-Attention)

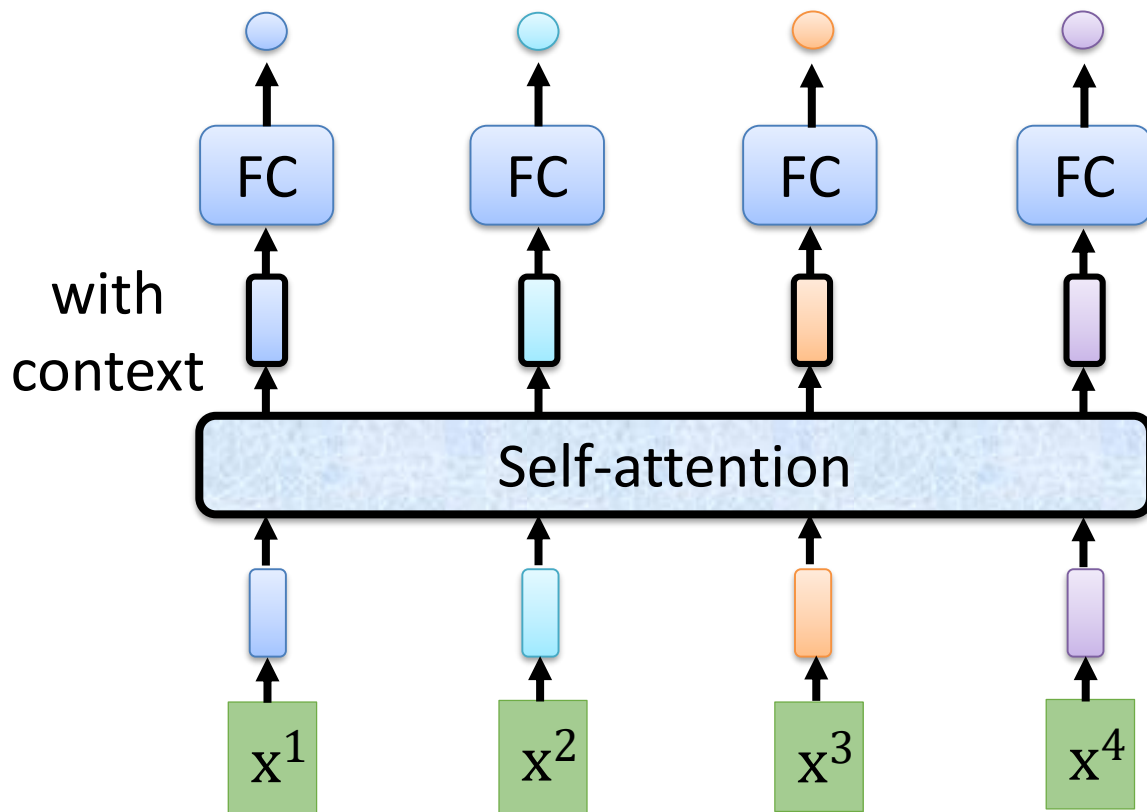
$$b^i = W^O \begin{bmatrix} b^{i,1} \\ b^{i,2} \end{bmatrix}$$





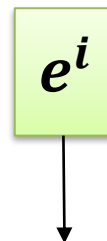
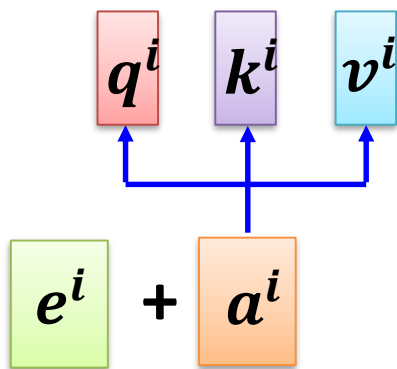
# 自注意力机制 (Self-Attention)

- 自注意力机制缺陷
  - 忽略了序列中的位置信息



# 自注意力机制 (Self-Attention)

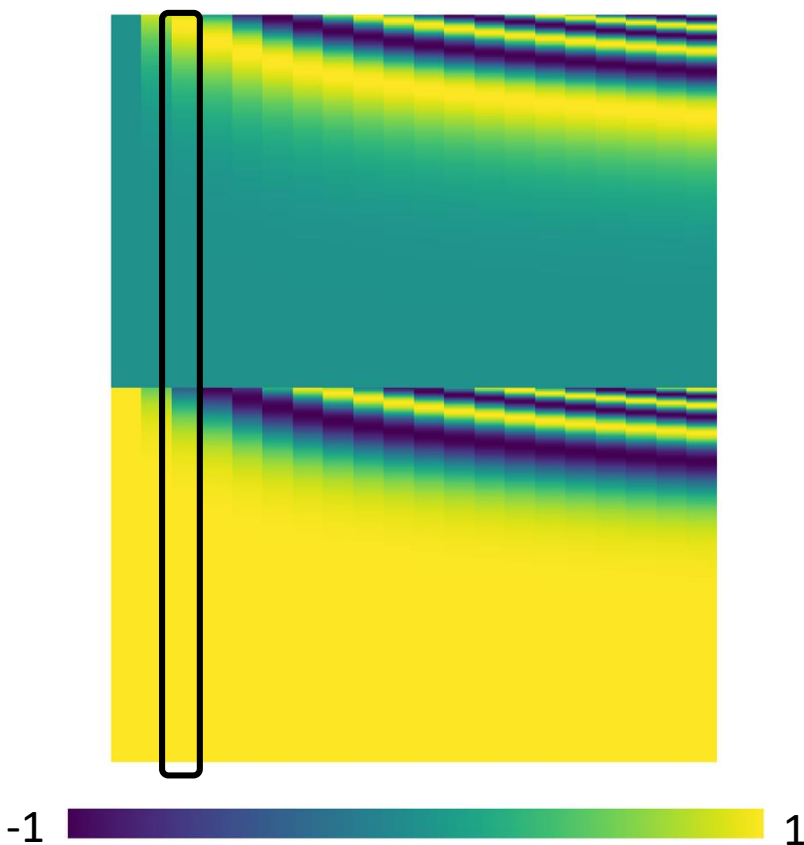
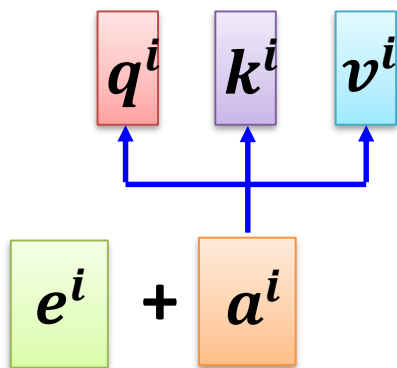
- 位置编码 (Positional Encoding)
  - 为每个位置引入一个位置编码  $e^i$ 
    - 人工构造
    - 参数学习



$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

# 自注意力机制 (Self-Attention)

- 位置编码 (Positional Encoding)
  - 为每个位置引入一个位置编码  $e^i$ 
    - 人工构造
    - 参数学习

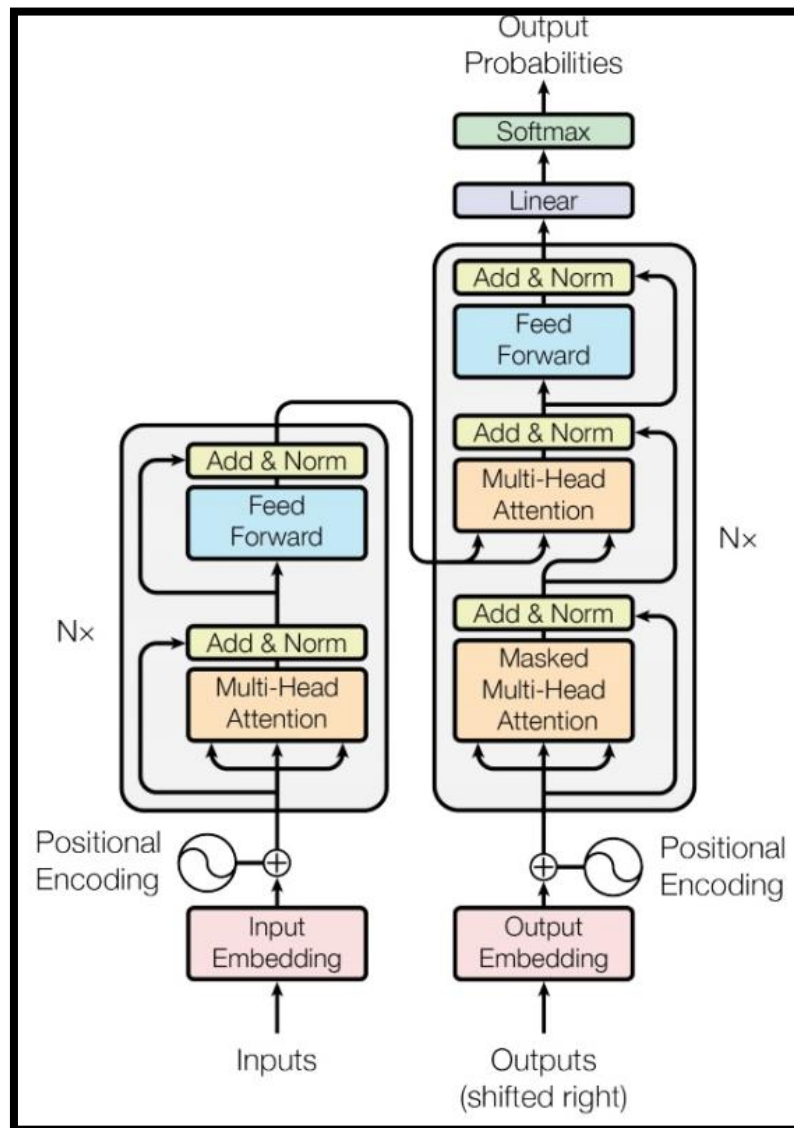


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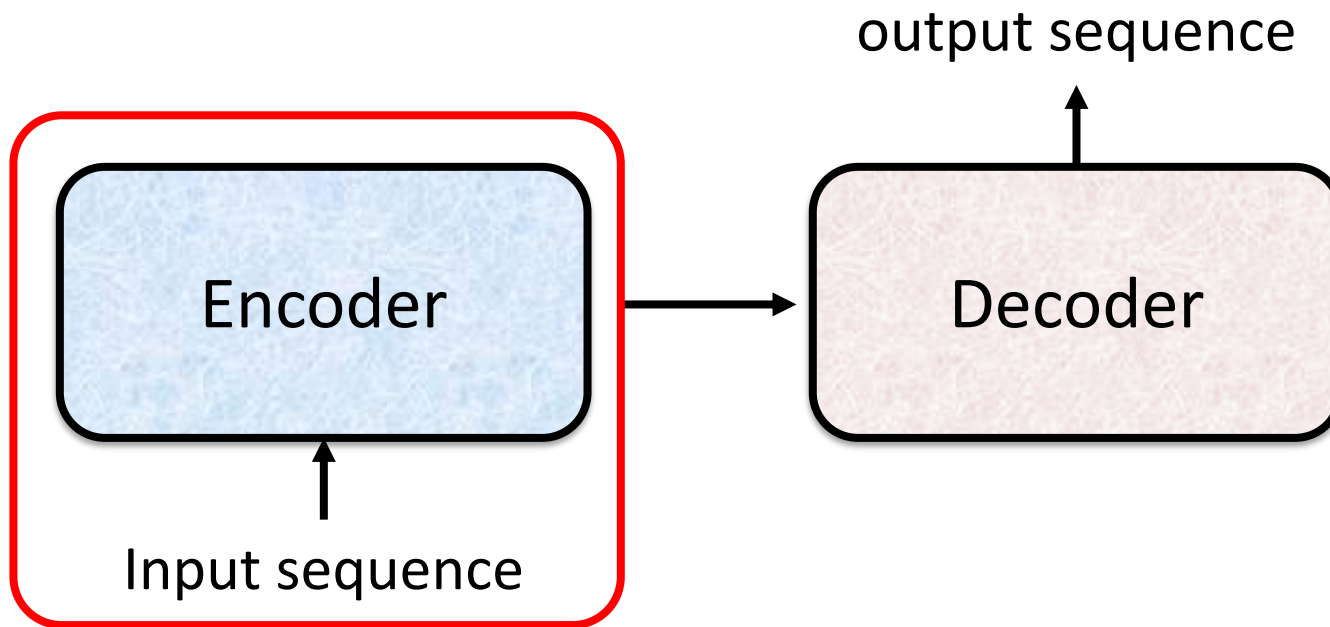
# Transformer

- 整体架构
  - Encoder-Decoder



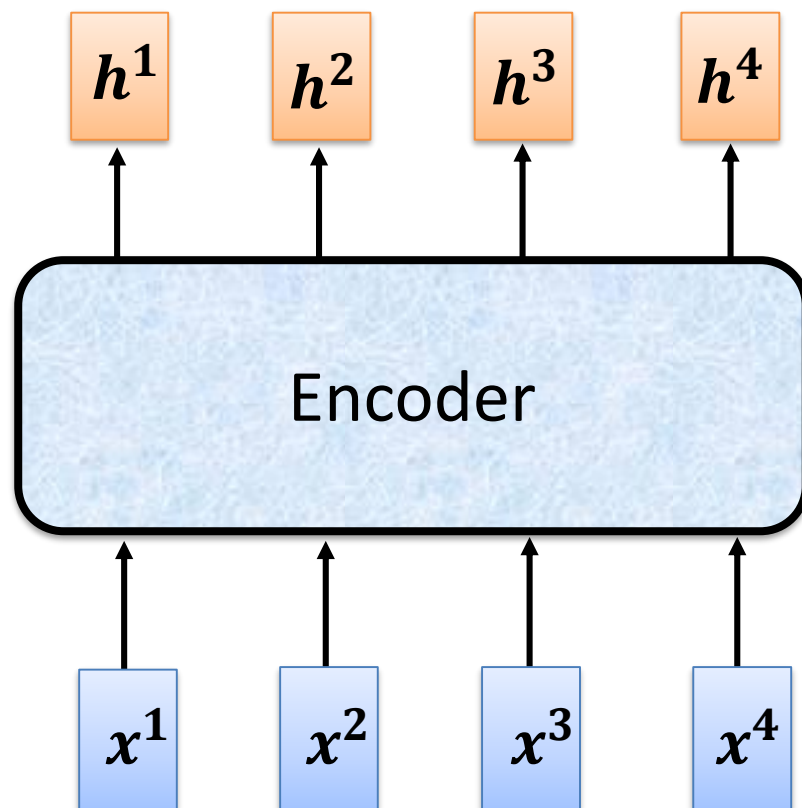
# Transformer

- 整体架构
  - Encoder-Decoder

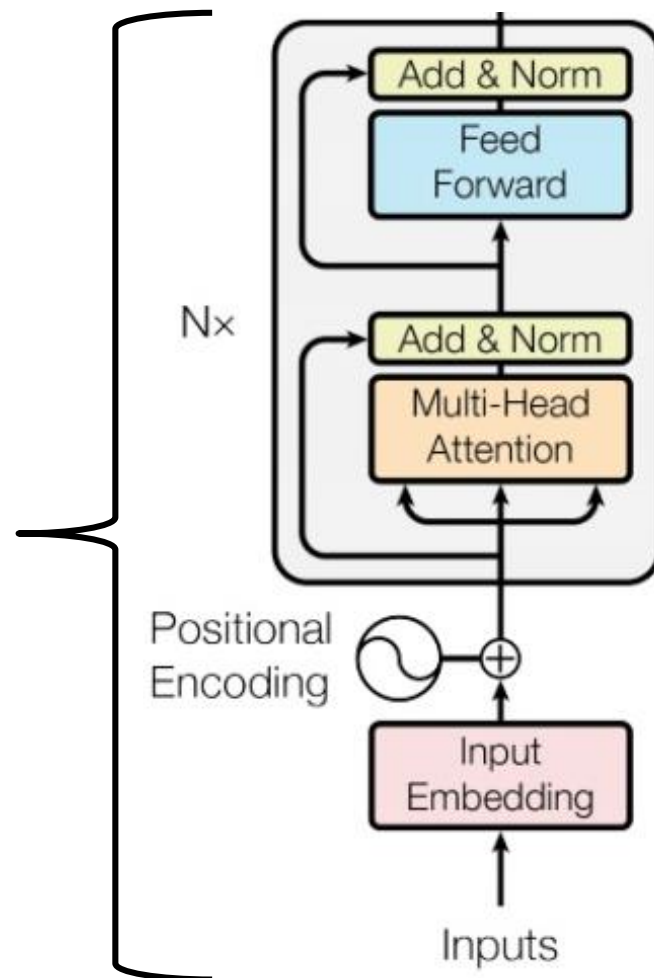


# Transformer

- 整体架构
  - Encoder

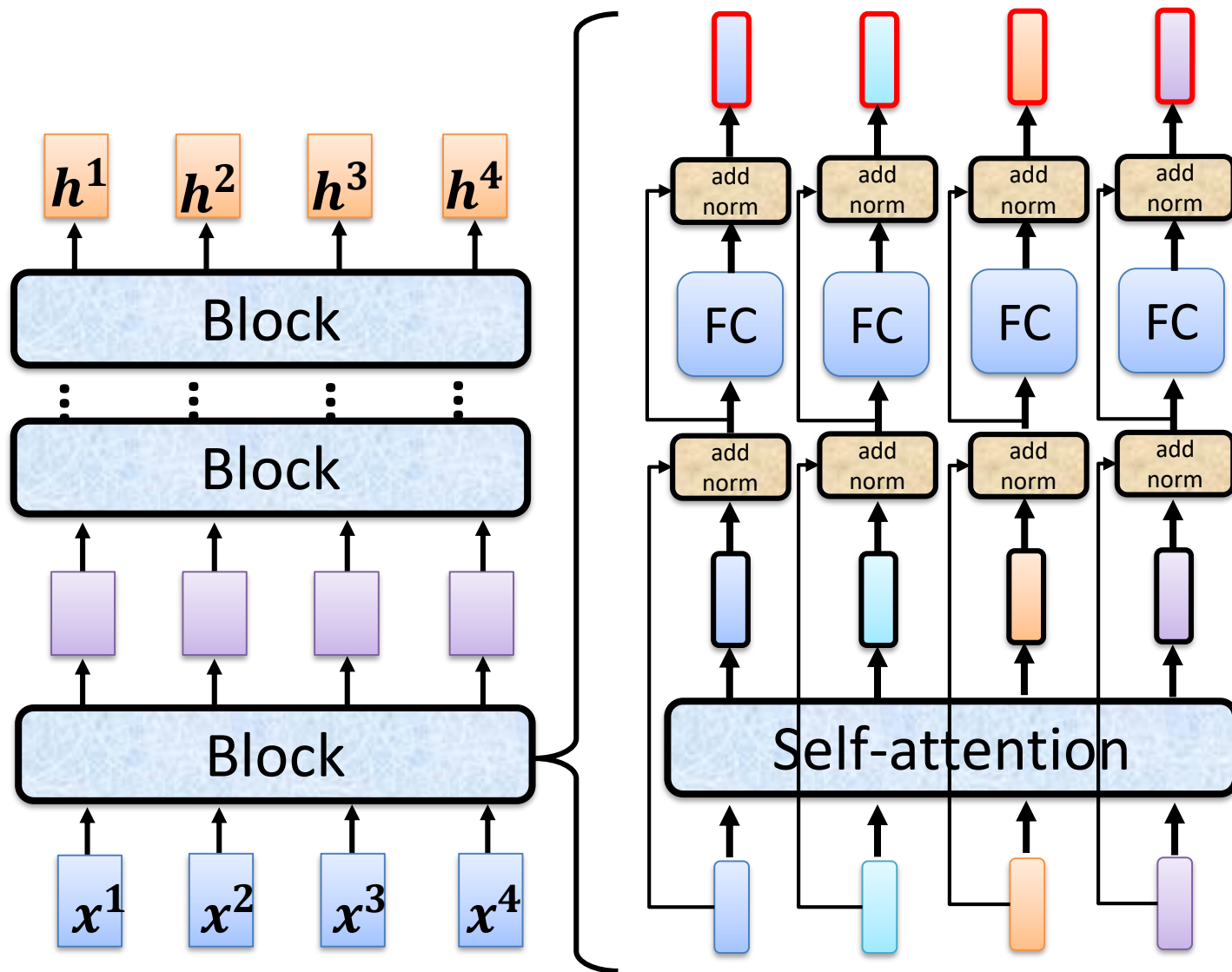


## Transformer's Encoder



# Transformer

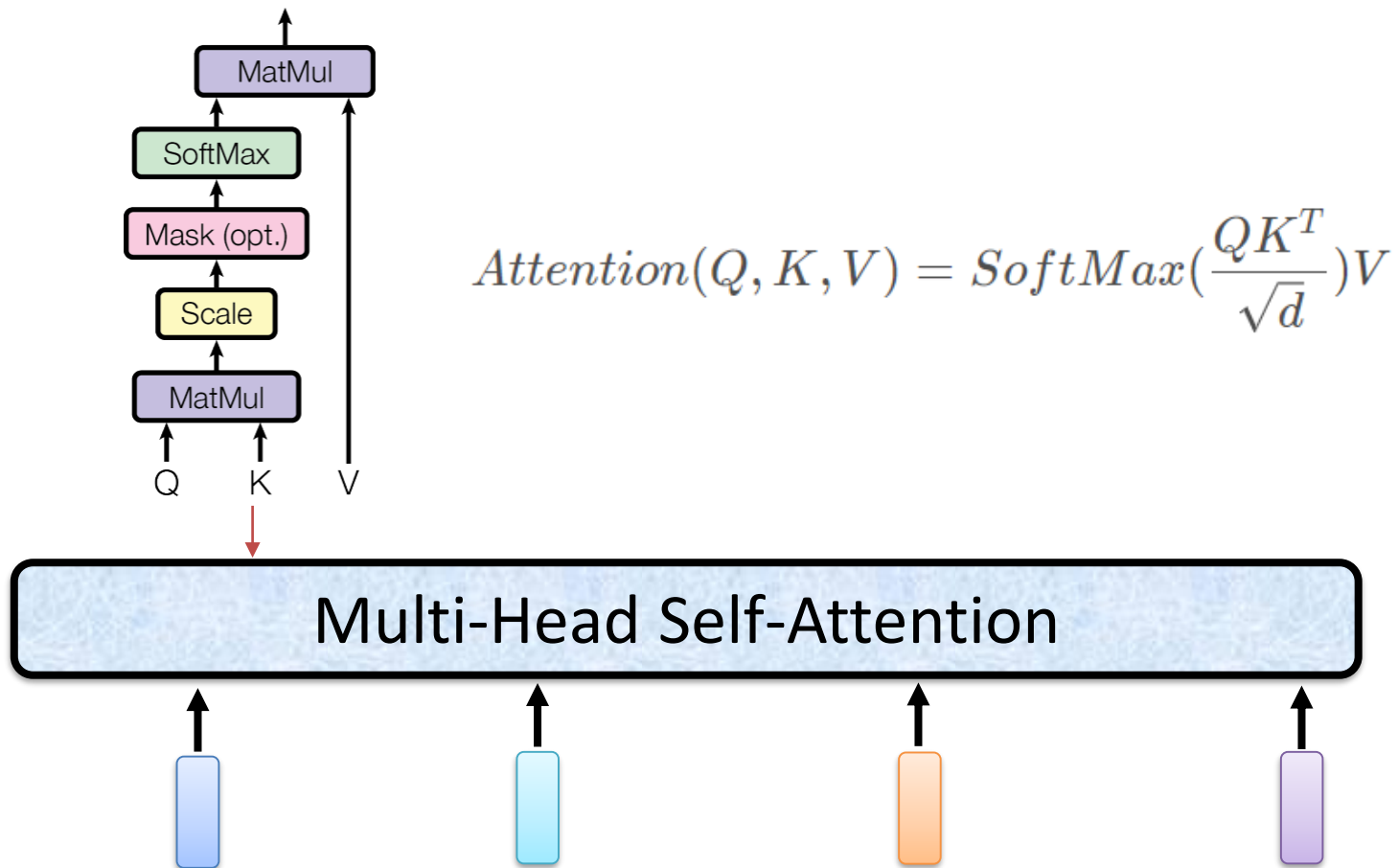
- Encoder





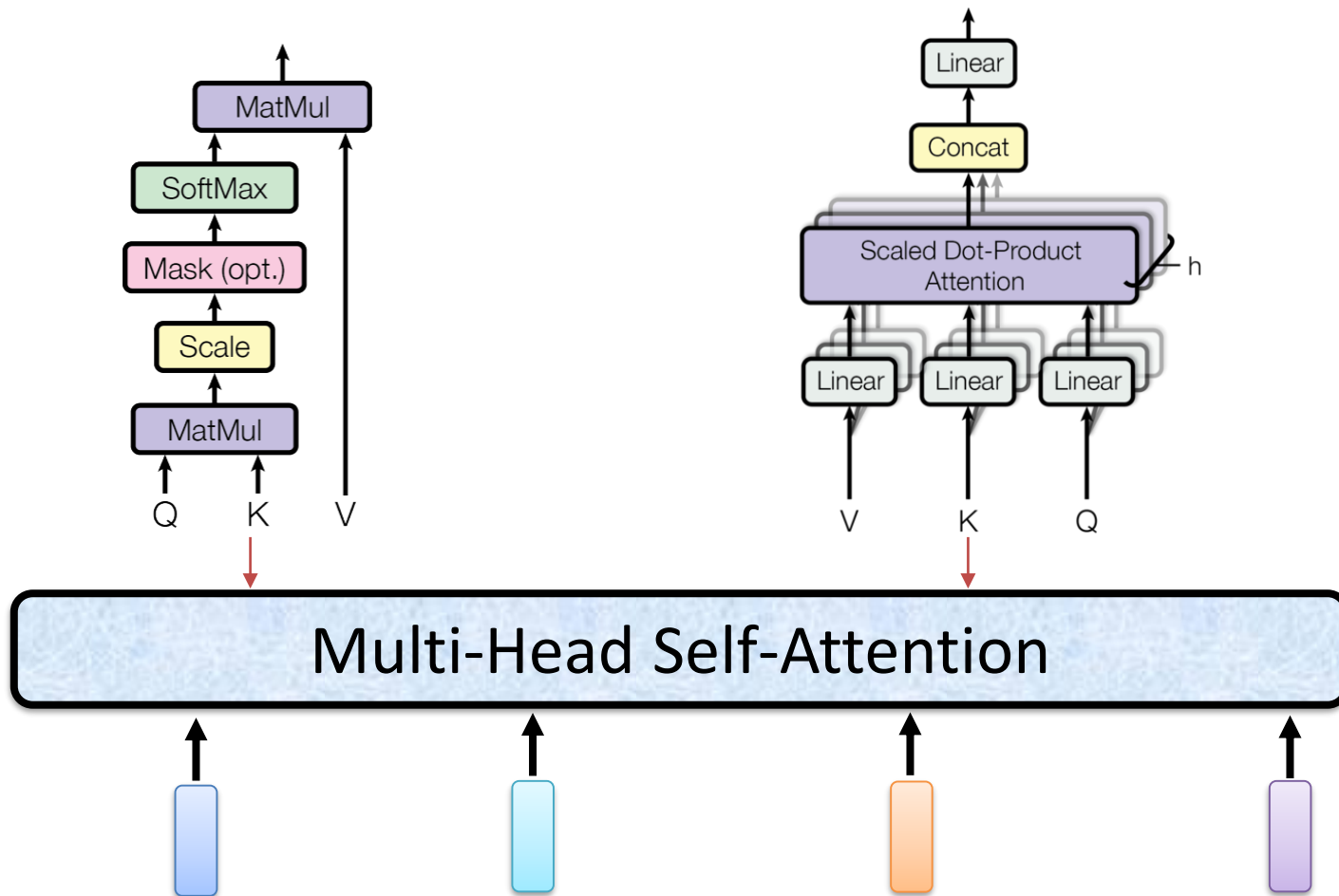
# Transformer

- Encoder
  - Scaled Dot-Product Attention



# Transformer

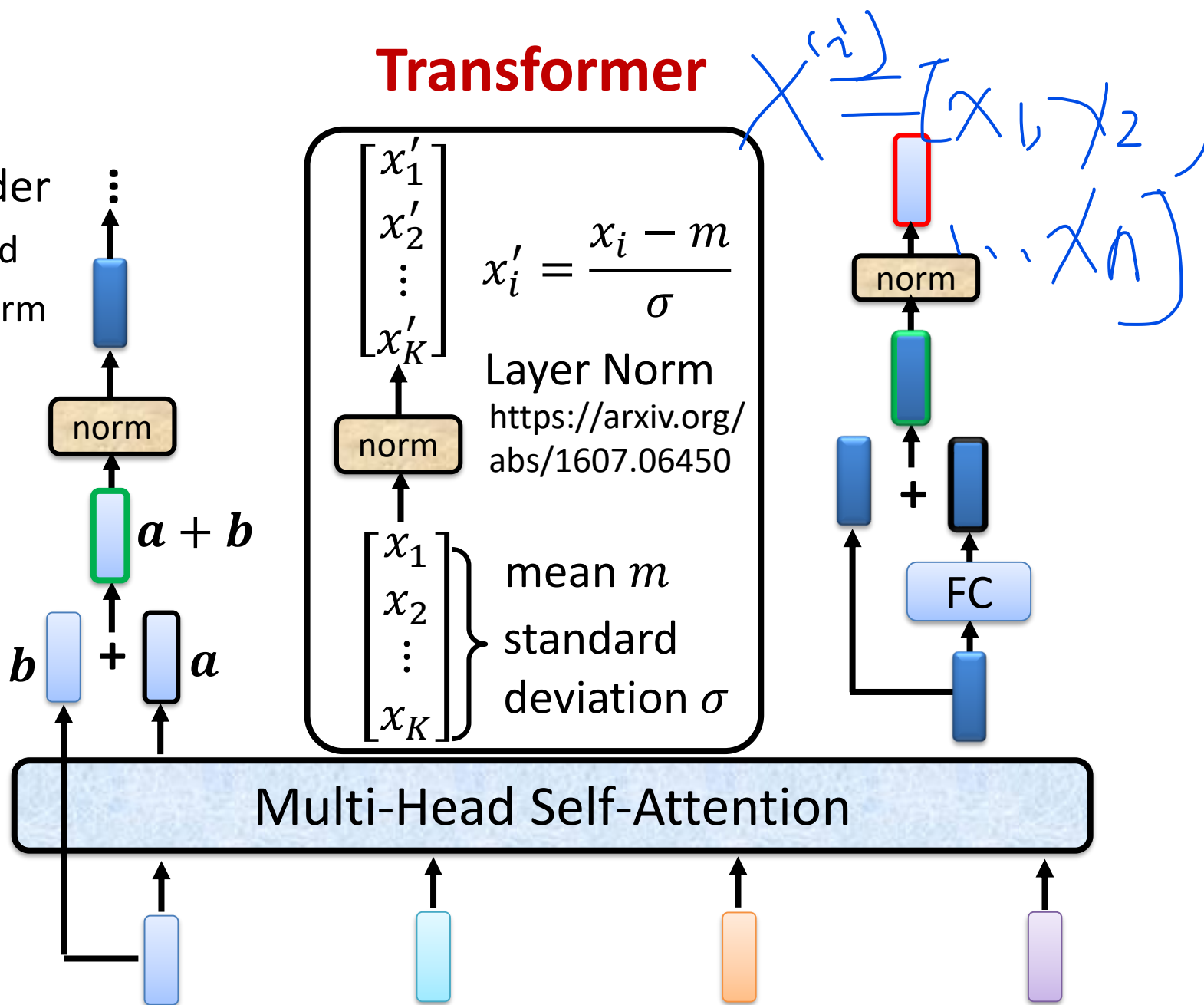
- Encoder
  - Scaled Dot-Product Attention



# Transformer

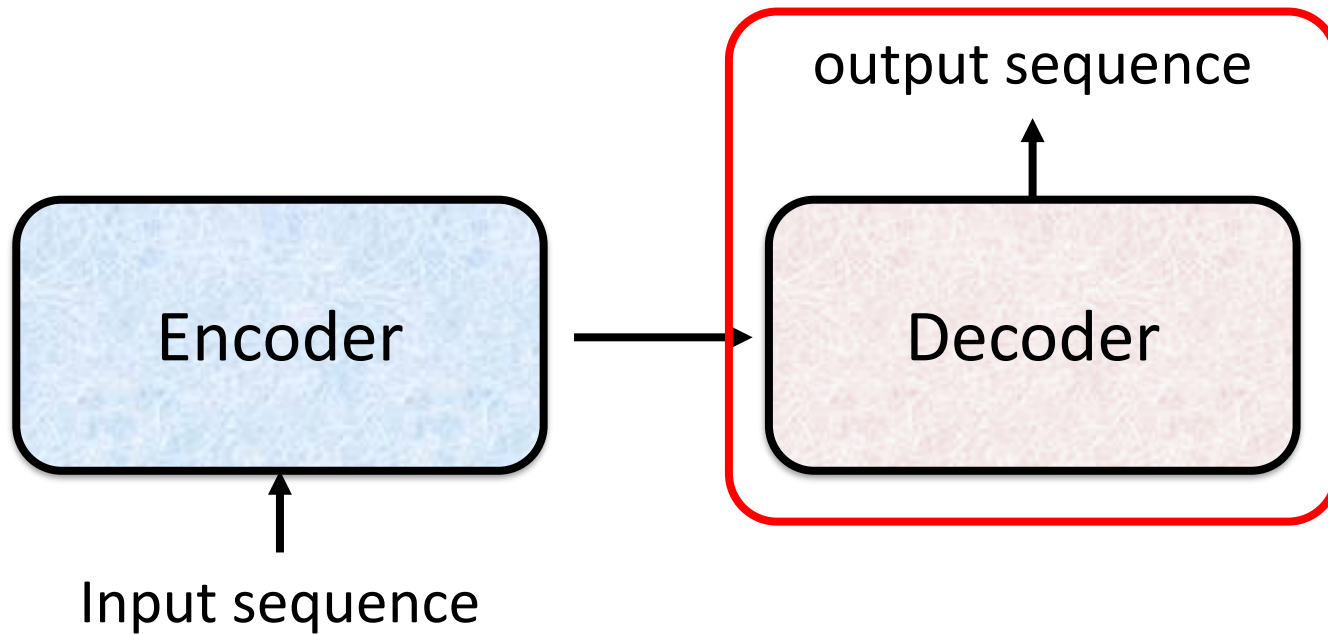
- Encoder

- Add
- Norm



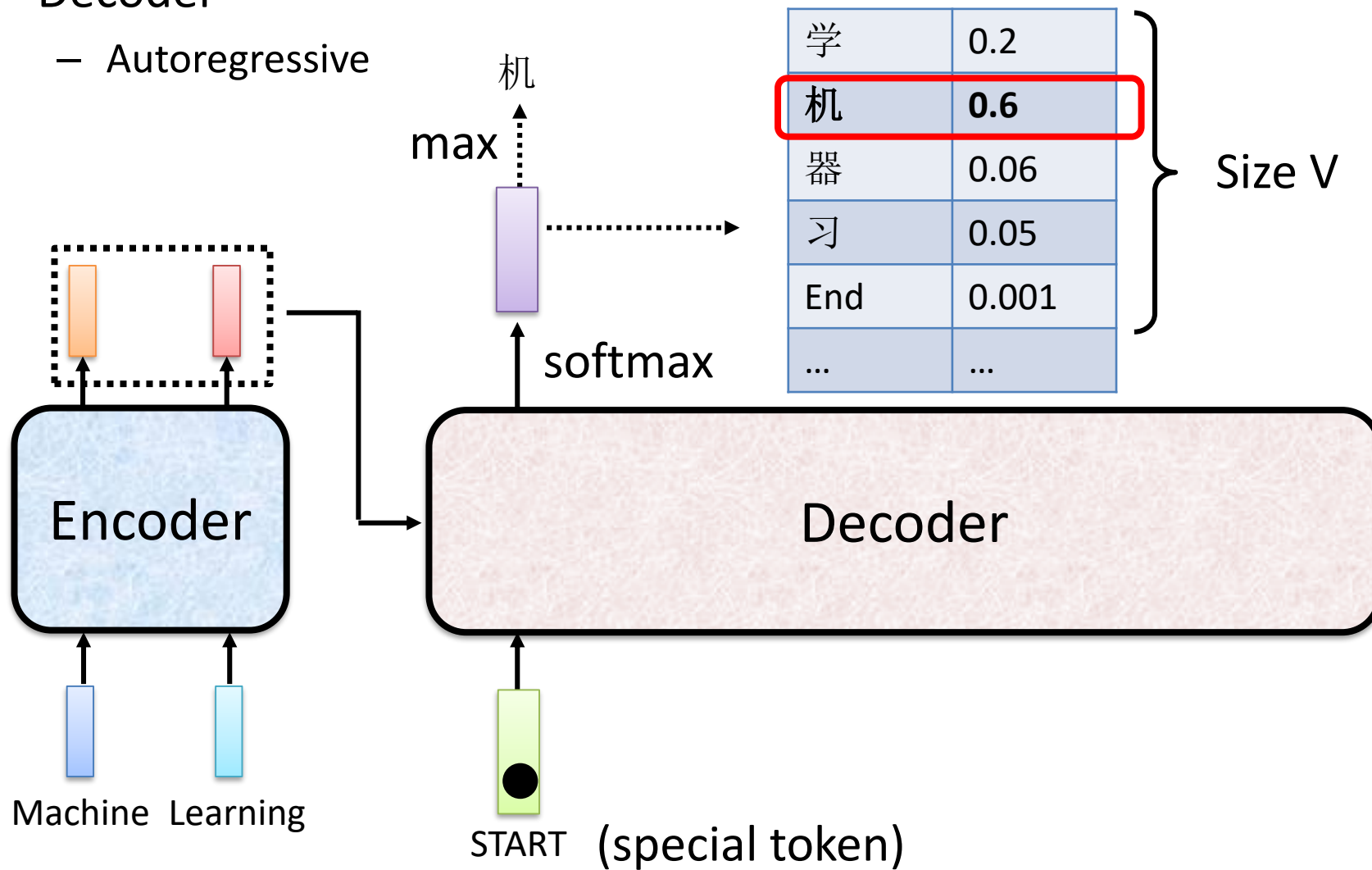
# Transformer

- Decoder
  - Autoregressive



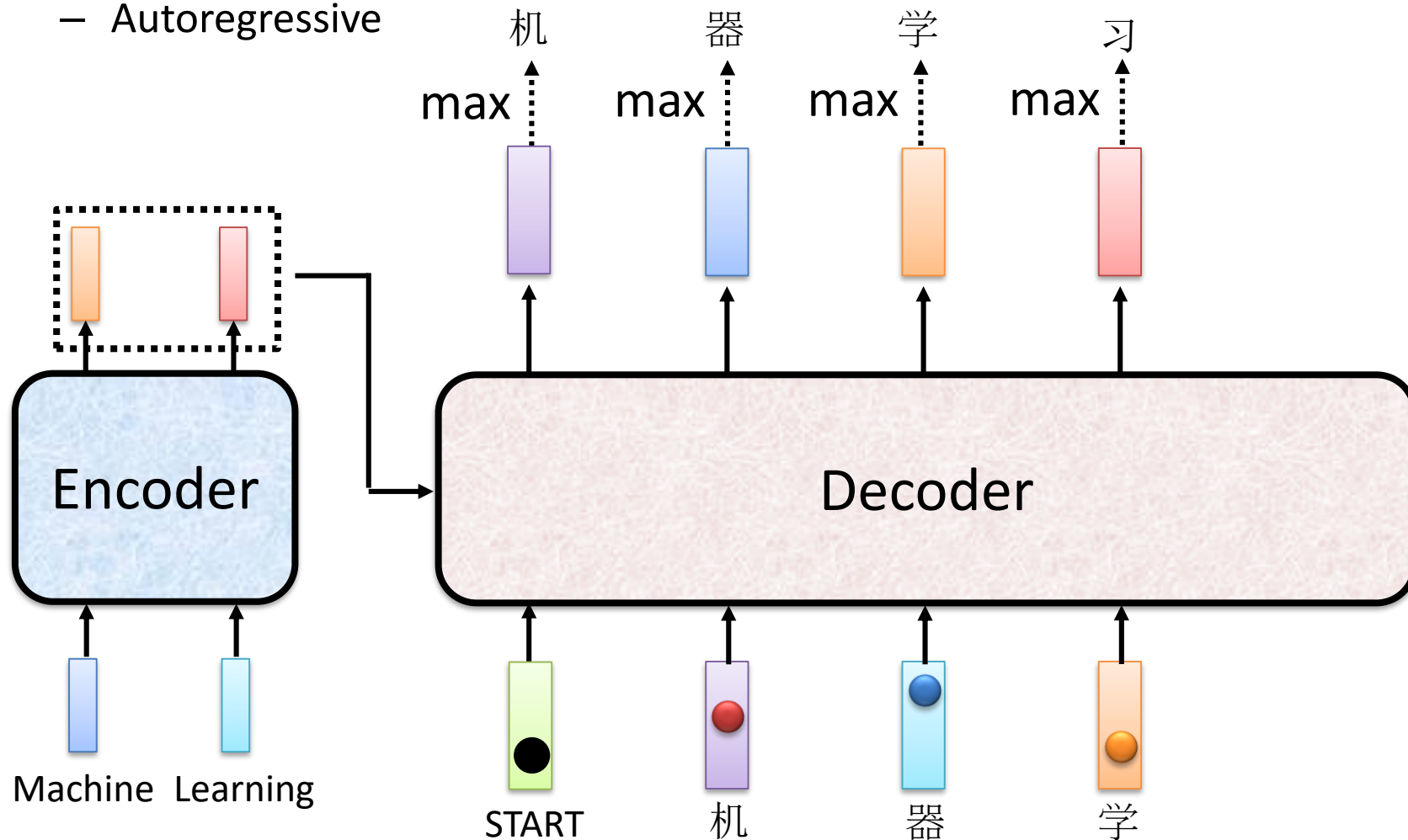
# Transformer

- Decoder
  - Autoregressive



# Transformer

- Decoder
  - Autoregressive

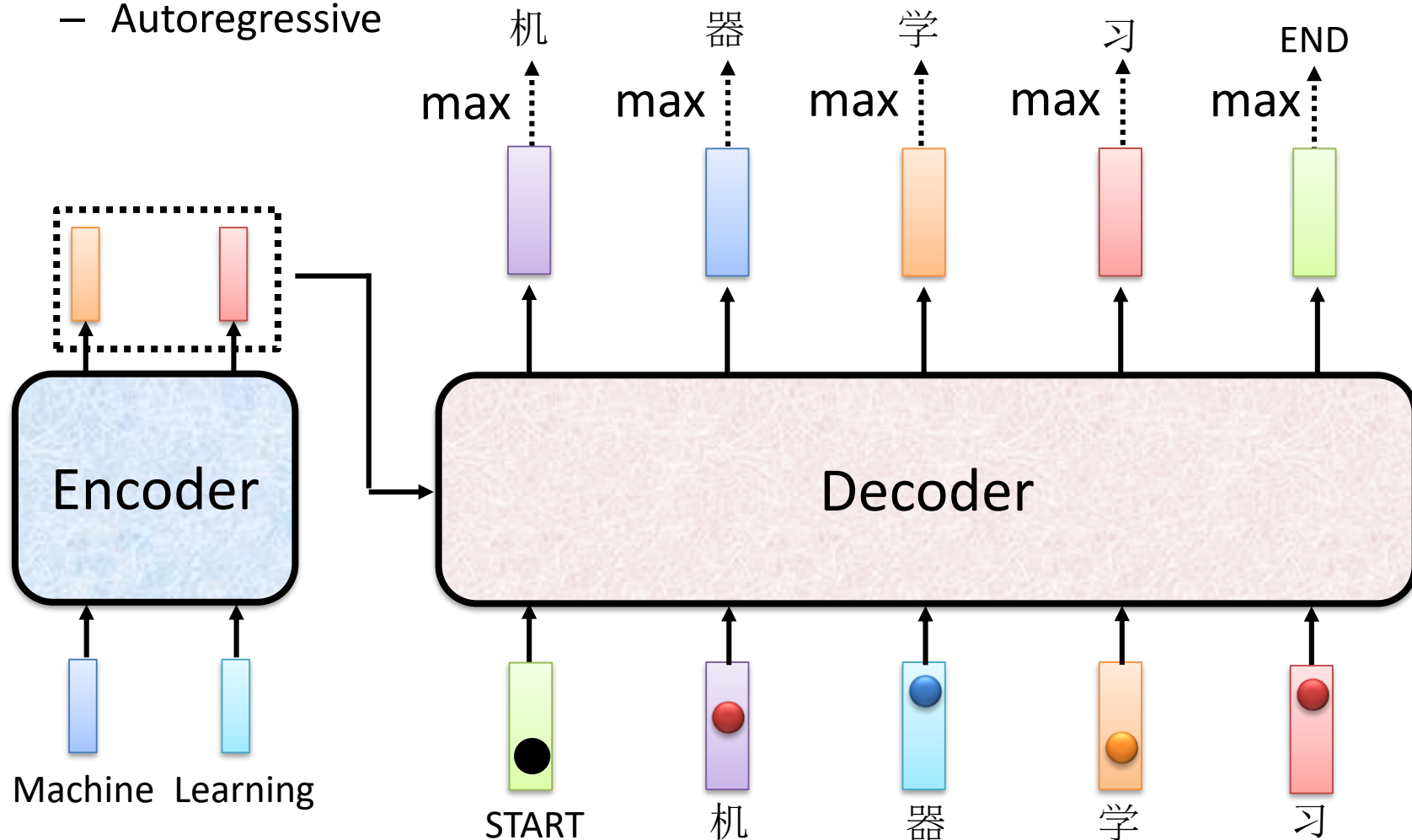


# Transformer

- Decoder
  - Autoregressive

|     |       |
|-----|-------|
| 学   | 0.001 |
| 机   | 0.01  |
| 器   | 0.001 |
| 习   | 0.001 |
| End | 0.9   |
| ... | ...   |

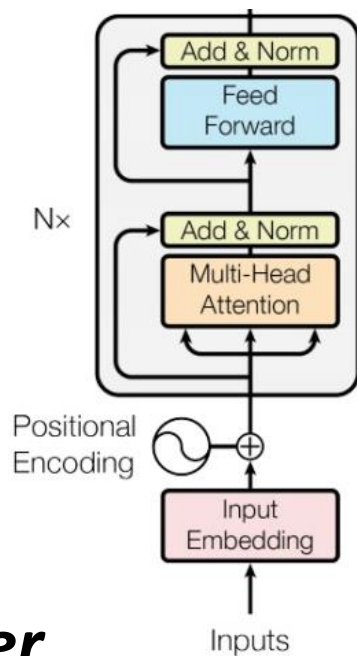
Size V



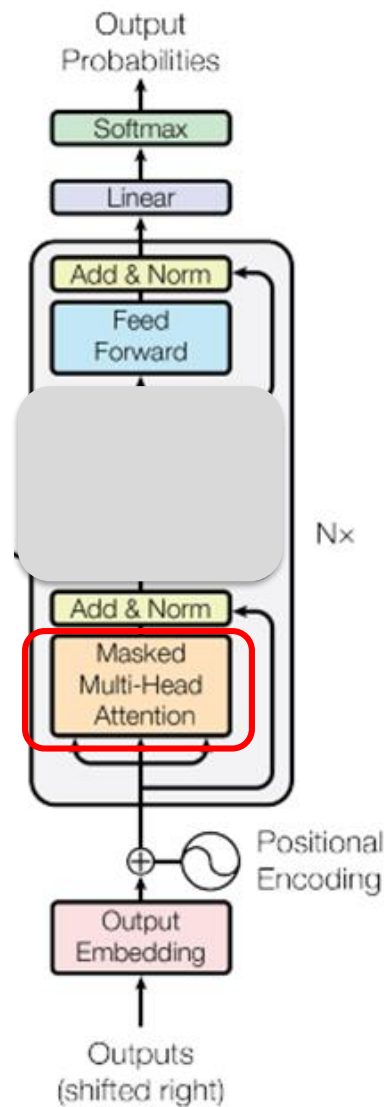
# Transformer

- Decoder
  - Autoregressive

**Encoder**



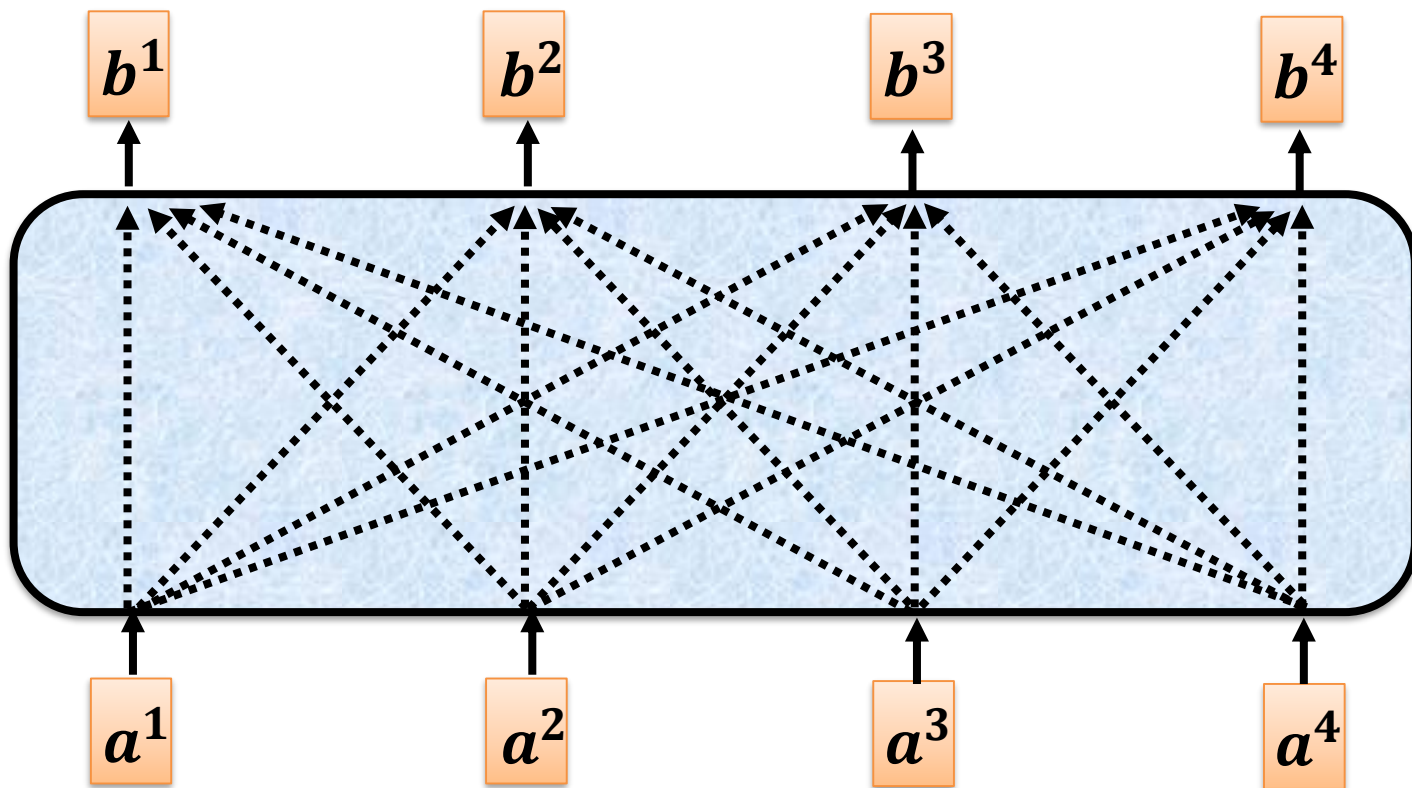
**Decoder**





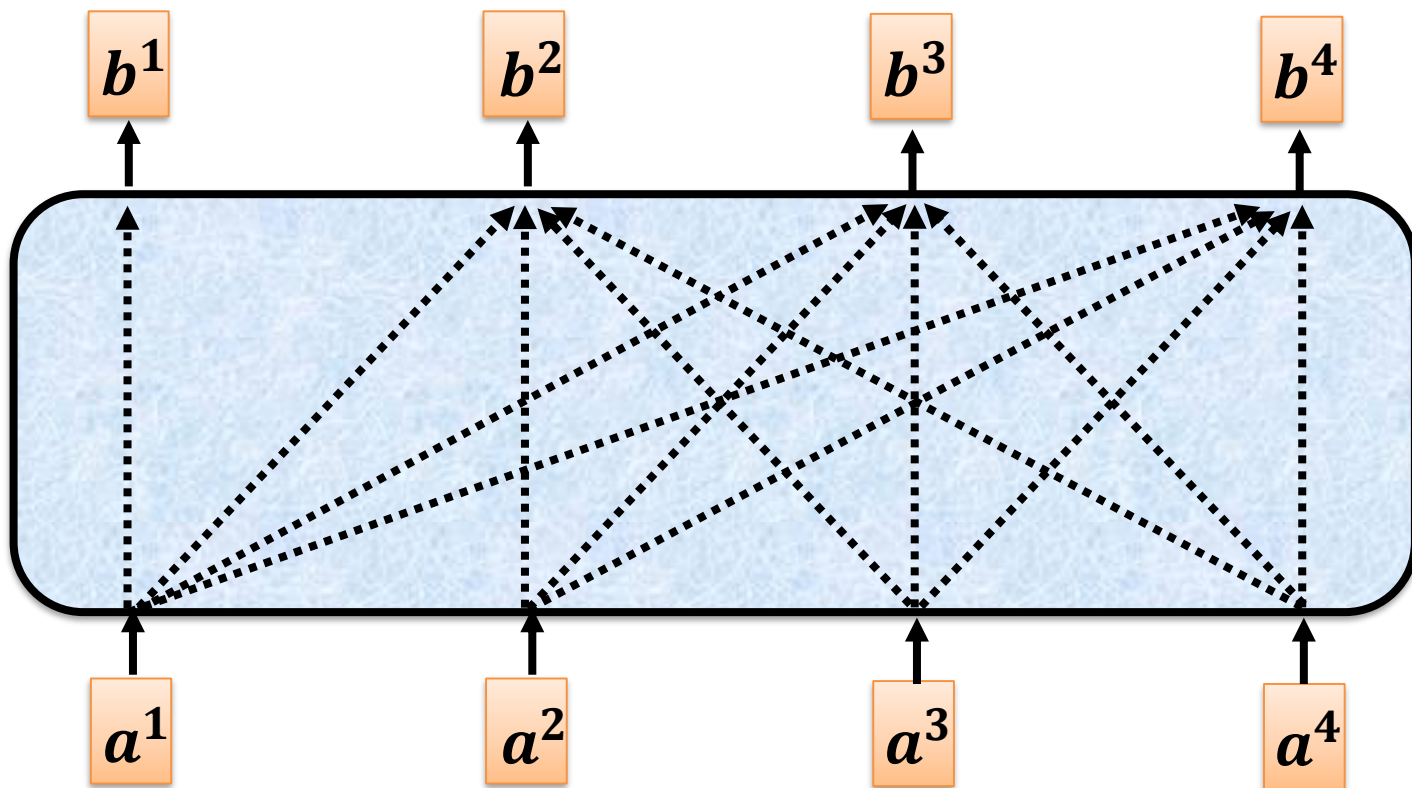
# Transformer

- Decoder
  - Self-Attention



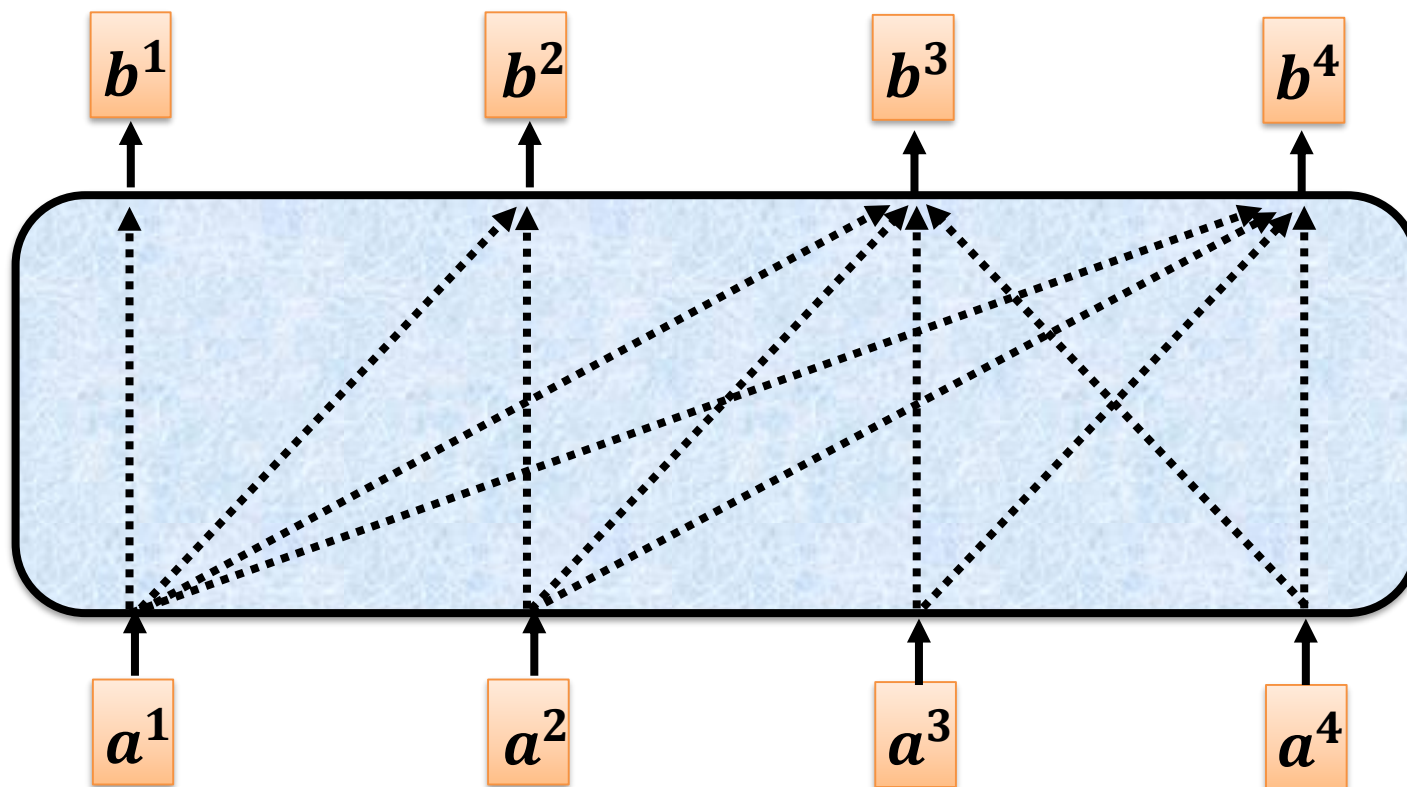
# Transformer

- Decoder
  - Masked Self-Attention



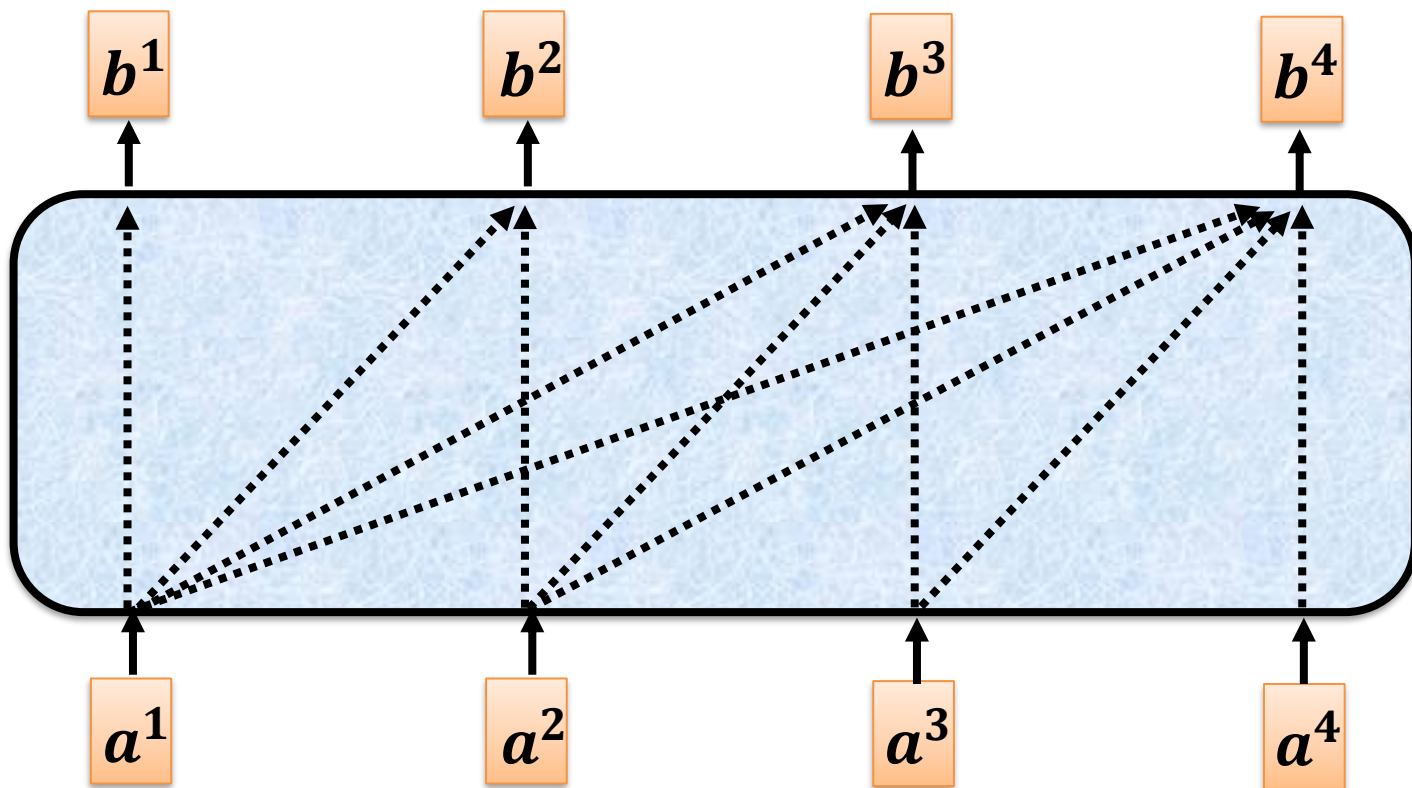
# Transformer

- Decoder
  - Masked Self-Attention



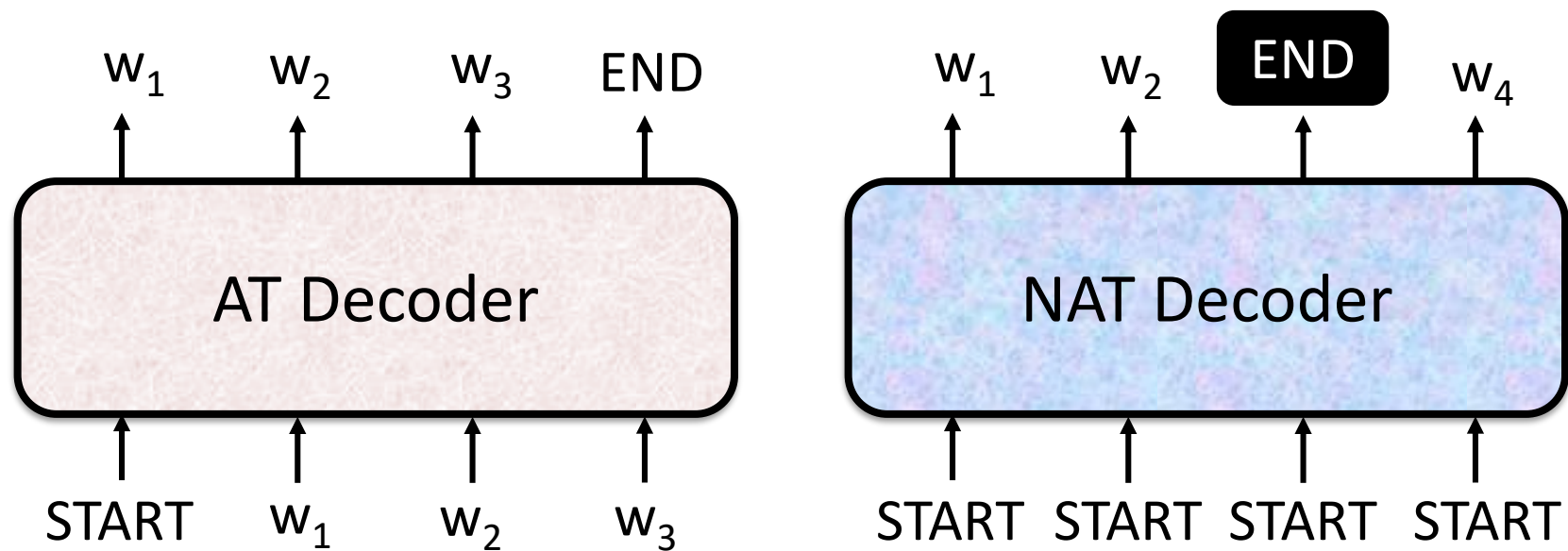
# Transformer

- Decoder
  - Masked Self-Attention



# Transformer

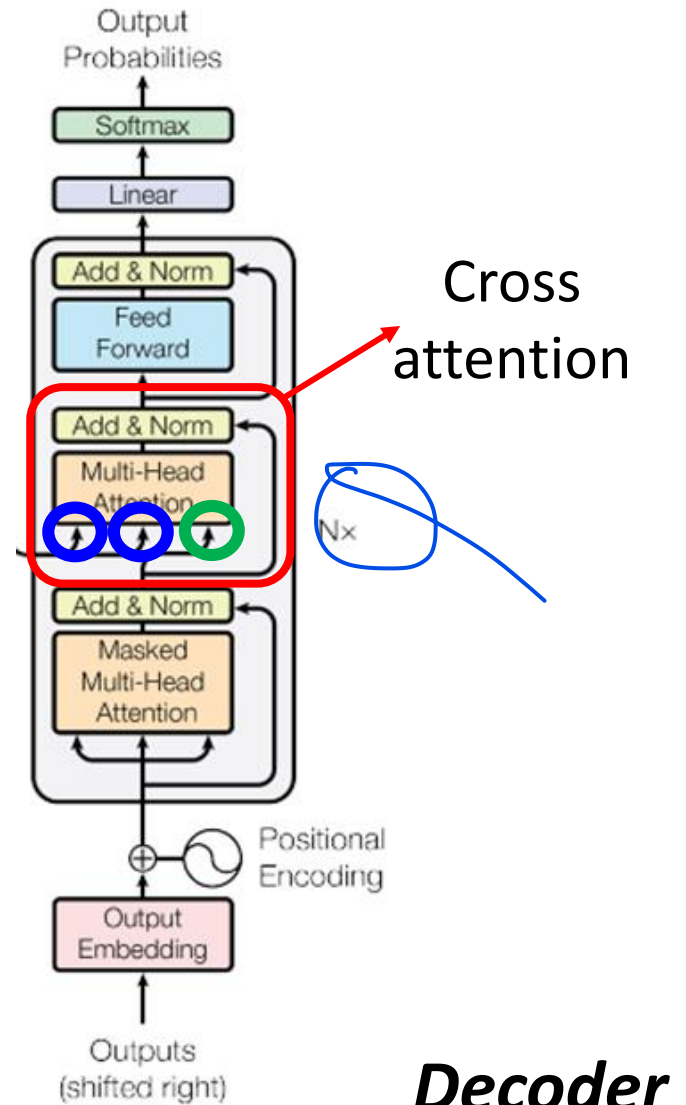
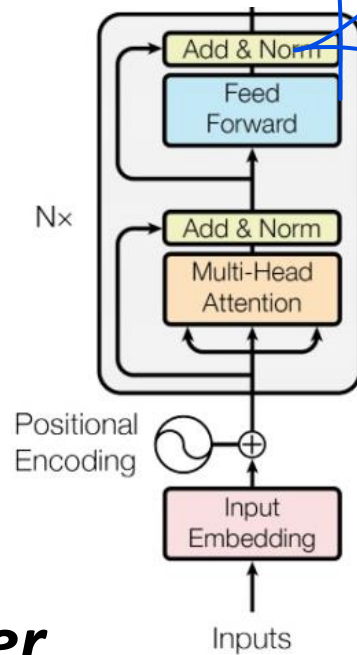
- Decoder
  - Autoregressive (AT) -> Non-Autoregressive (NAT)



# Transformer

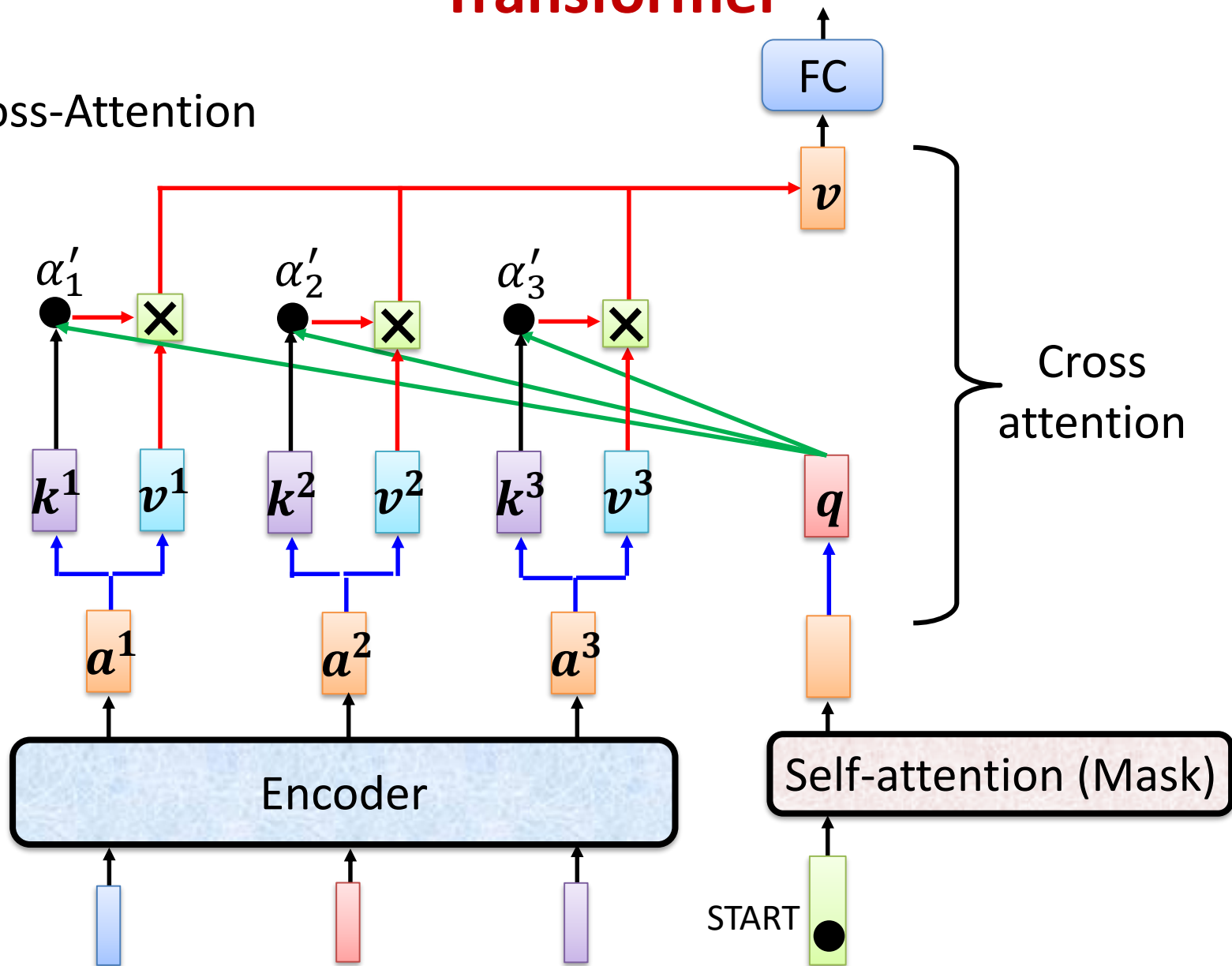
- Decoder
  - Cross-Attention

Encoder



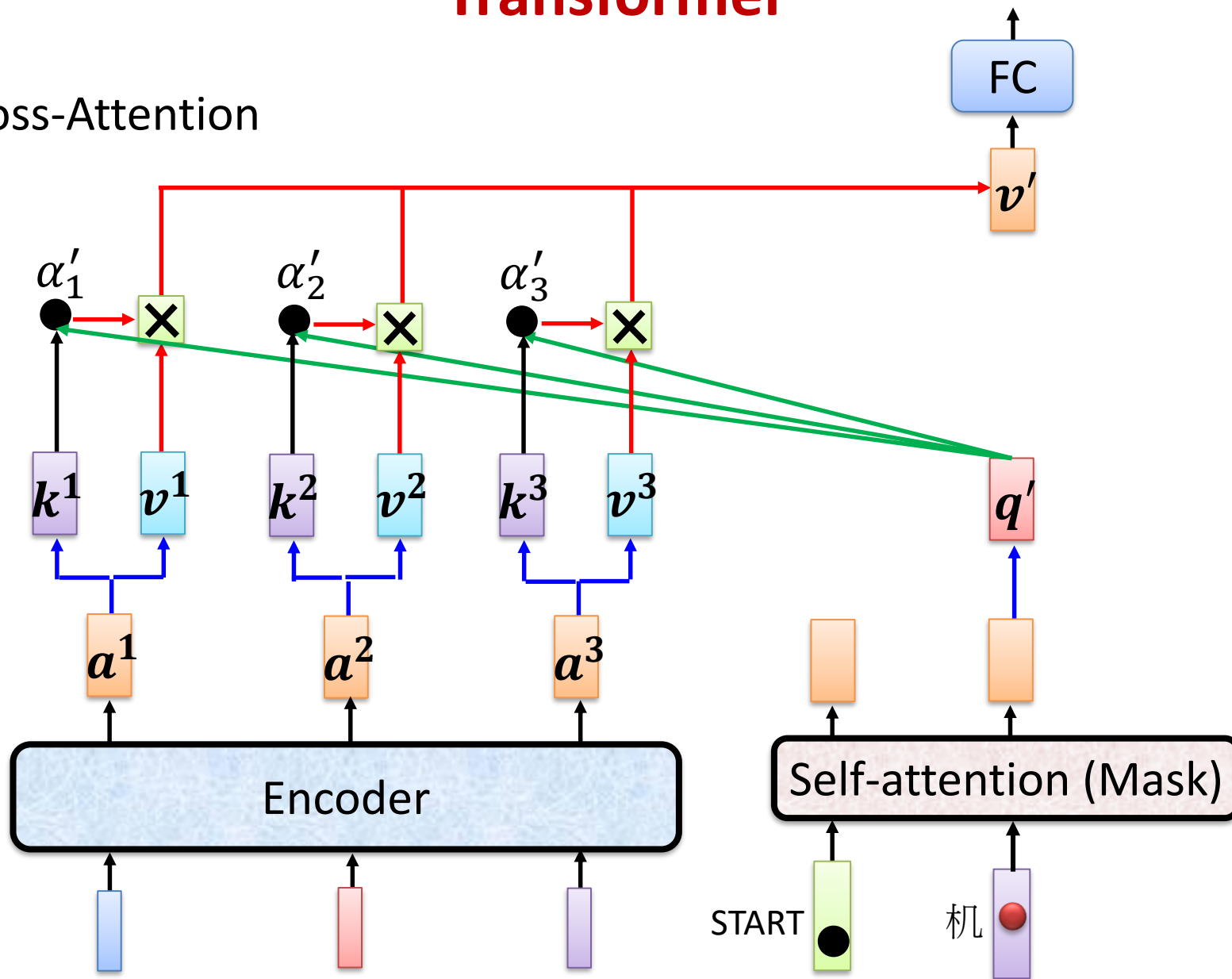
# Transformer

- Cross-Attention



# Transformer

- Cross-Attention

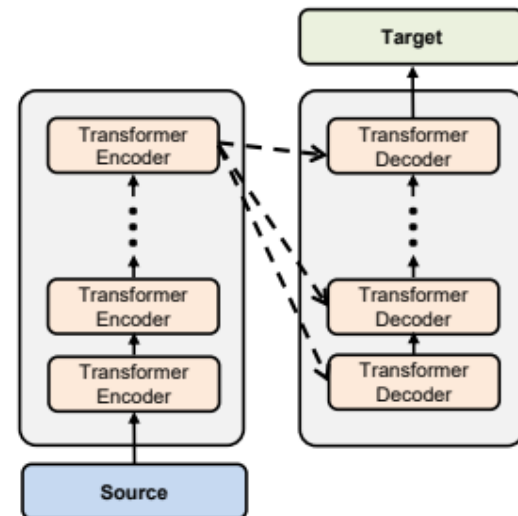




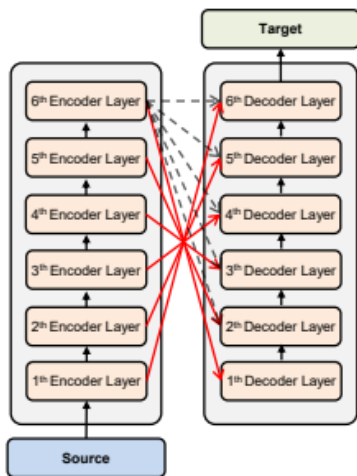
# Transformer

- Cross-Attention

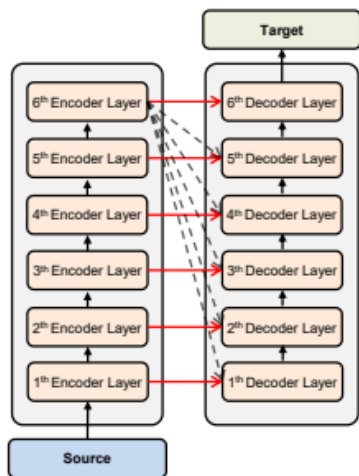
- 图片来源: <https://arxiv.org/abs/2005.08081>



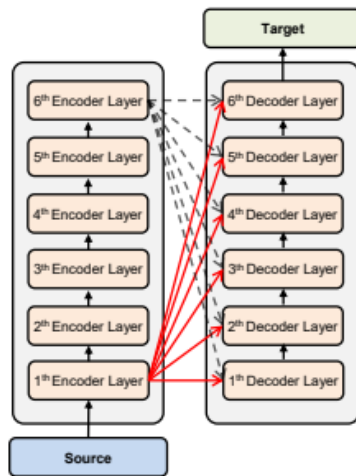
(a) Conventional Transformer



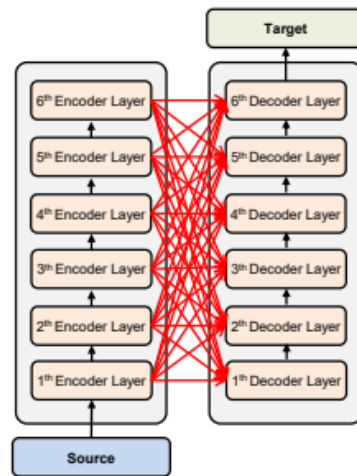
(a) Granularity Consistent Attention



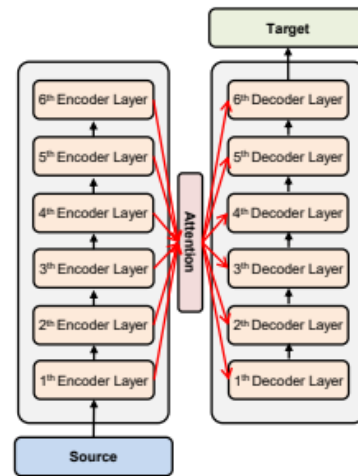
(b) Granularity Parallel Attention



(c) Fine-Grained Attention



(d) Full Matching Attention



(e) Adaptive Matching Attention

# 课程作业

- 基于Transformer的机器翻译系统
  - 问题描述
    - 利用Transformer，将输入的英文句翻译成中文
  - 数据集
    - 输入：一句英文（e.g. tom is a student .）
    - 输出：中文翻译（e.g. 汤姆 是个 学生 。）
    - 训练集：18000句
    - 验证集：500句
    - 测试集：2636句
  - 数据集下载地址
    - 链接：[https://pan.baidu.com/s/1Vb3PvFkfCvJ\\_JdapgEU\\_Hg](https://pan.baidu.com/s/1Vb3PvFkfCvJ_JdapgEU_Hg)
    - 提取码：h43i
  - 要求
    - Tensorflow或者Pytorch实现

# 课程作业

- 基于Transformer的机器翻译系统
  - 评价指标
    - BLEU score
      - BiLingual Evaluation Understudy, IBM
      - 将机器翻译产生的候选译文与人翻译的多个参考译文相比较，越接近，候选译文的正确率越高。
      - 统计同时出现在系统译文和参考译文中的n元词的个数，最后把匹配到的n元词的数目除以系统译文的n元词数目，得到评测结果。

# 课程作业

- 基于Transformer的机器翻译系统
  - 评价指标

$$BLEU = BP \times \exp\left(\sum_{n=1}^N w_n \log p_n\right)$$

长度过短句子的  
惩罚因子

$$w_n = 1/N$$

最大语法的阶  
数，实际取4。

出现在答案译文中的  
 $n$ 元词语接续组占候  
选译文中 $n$ 元词语接  
续组总数的比例。

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

$c$ 为候选译文中单词的个数， $r$ 为答案  
译文中与 $c$ 最接近的译文单词个数。

**BLEU 分值范围：0 ~ 1，分值越高表示译文质量越好，分值  
越小，译文质量越差。**

# 课程作业

- 基于Transformer的机器翻译系统

- 参考资料

- 《神经网络与深度学习》 第15.4.2小节
    - <http://nlp.seas.harvard.edu/2018/04/03/attention.html>
    - Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I.. Attention is all you need. In NIPS 2017.
    - Wang, Q., Li, B., Xiao, T., Zhu, J., Li, C., Wong, D. F., & Chao, L. S. (2019, July). Learning Deep Transformer Models for Machine Translation. In ACL 2019.



**欢迎提问！**